Multidimensional Goodhart

How measurement reshapes the terrain under measurement

Working draft — Intro and Chapters 1–5

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1 Intro: Working spine

1.1 The empirical phenomenon

The empirical phenomenon I want this paper to help science model is not just "the proxy stops working". Scalar Goodhart already says that. The phenomenon is: optimization pressure reshapes the residual error landscape. Once a measure is used for control, the error between the proxy and the target gets a structure: direction, support, curvature, tail behavior, exchange rates, and response dynamics. If the system is multidimensional, that structure is the interesting object. The proxy can fail by drifting along hidden goal coordinates, by changing the coupling between visible and hidden variables, by recruiting new agents into gaming, by moving from one cheap action channel to another, or by changing which residuals remain legible to the evaluator.

The licensed claims in this draft are deliberately narrower than that phenomenon. In a pure selection channel, hidden drift is a baseline-response object and admits two Cauchy–Schwarz bounds: a coordinate-explicit Euclidean bound after hidden coordinates are declared, and a declared value/operator bound after the value metric is declared. In a toy intervention channel, quadratic Stackelberg gaming gives a wedge of width $\sqrt{2\kappa V}$, while the convex score-deficit proposition gives a private-cost budget only after action geometry is declared. In multidimensional scorecards, additive and conjunctive aggregation have different signs, and additive fixed-deficit conservation holds only under the harm-per-score exchange-rate condition. In response-shape claims, quadratic costs, fixed charges, caps, low-rank affordances, and search priors are guardrails on different models, not evidence for a generic complexity attractor. For applications, the transfer rule is the response-modeling contract: declare type, response channel, action/search geometry, aggregation, hidden harm, and falsifiers before importing any calculation.

The obvious objection is that this could become a way of naming the residue after the fact. If the story is "there were many dimensions, something bad happened in one of them, and now we call it multidimensional Goodhart", then that is not a theory. The point of the framework has to be stronger and more brittle: before seeing the next failure, it should say which response model is being claimed and what shape of distortion that model predicts.

1.2 Why this approach

I do not think the right claim is "more dimensions means more Goodhart". That failed almost immediately. In a pure selection model, if hidden coordinates are independent of the selected proxy, thresholding does not move them. Even when they move, the dimension count enters through a coupling or variance budget, not by itself. The elementary intuition is worth keeping visible: an added measured coordinate can be positively associated with a hidden value coordinate, negatively associated with it, or approximately unrelated to it; under selection these three cases can respectively move the selected population toward that hidden value, away from it, or not much at all. This is obvious as a matter of correlation and conditional expectation, but it is one of the main reasons dimension count alone cannot carry the theory. I also do not think the right claim is "Goodhart pressure makes residual error more complex". Selection channels do not optimize residuals at all; they reweight a baseline. Intervention channels optimize according to cost, search, caps, fixed charges, and aggregation. Complexity only becomes predictive after the response geometry and the complexity measure are named in advance.

The narrower motivation is that scalar Goodhart is often doing two different compressions at once. First, it compresses the target into a single number. Second, it compresses the residual error into a single loss, usually without declaring the metric that says which directions of error are comparable. That is not wrong when the scalar value functional is genuinely part of the model: a declared $v \cdot H$, utility function, or loss norm is exactly what Chapter 2′s operator bound asks for. But if the scalar is only a reporting convenience, it can make incomparable errors look interchangeable. A proxy can improve in the reported scalar while moving along a hidden direction that the scalarization has priced at zero, or while trading a small loss on a high-value coordinate for a large gain on a low-value coordinate because the weights were inherited from the dashboard rather than from the goal.

The opposite repair, running scalar Goodhart separately on each coordinate, also loses the object of interest. Coordinate-wise analyses can tell us whether each proxy component becomes noisy, biased, or gameable on its own. They do not by themselves say how the error vector moves: whether gaming substitutes across dimensions, whether a closed channel spills pressure into another, whether a conjunctive gate changes the active constraint set, or whether a Pareto rule preserves diversity that a weighted sum would collapse. Those are geometric questions about the response channel and aggregation rule. The multidimensional framework is motivated by this gap: not by the claim that scalar models are false, but by the claim that scalar models silently choose, or omit, the metric that makes the residual shape intelligible.

The specific approach here is to make proxy-induced distortion modelable. That requires more objects than scalar Goodhart usually keeps visible. A model has to say whether pressure only reweights fixed states or changes behavior at fixed type. It has to say what the hidden coordinates are, how proxy components are aggregated, what actions are feasible, what they cost, what hidden harm functional is being protected, and what evidence would distinguish the proposed response channel from nearby alternatives. Without those declarations, "the metric broke" is too underspecified to imply a prediction.

This is why the selection/intervention split matters. In a selection channel, the policy changes weights over a fixed baseline distribution. Then the natural tools are statistical: covariance as local velocity, threshold response curves, weighted selection response, and reweighting bounds. In an intervention channel, the policy changes the response kernel at fixed type. Then the natural tools are geometric and economic: action spaces, cost functions, caps, fixed charges, stakes, feasible sets, and support functions. The same observed score movement can be selection in one declared type space and intervention in another, so the type/action representation is not a nuisance caveat. It is where the empirical claim enters.

1.3 What looks promising

The promising part is that several pieces already have the right shape. For selection, the framework gives response functionals of the baseline distribution. Covariance is not promoted into a universal primitive; it is the local derivative of hidden drift under infinitesimal Boltzmann selection only on the finite-mgf domain where that path exists and is differentiable. Threshold response handles hard cutoffs. Weighted response handles generic reweighting. The chi-square selection bound says that, once the hidden coordinates and norm are declared, bounded reweighting plus bounded baseline hidden variance implies bounded hidden drift. That is not a grand theorem, but it is exactly the kind of baseline-only statement selection deserves.

For intervention, the promising part is that the controlling quantities are not baseline statistics. In the quadratic Stackelberg toy, the gaming wedge is $\sqrt{2\kappa V}$: ease of gaming times stakes, not a property of last year's joint distribution. In the convex sketch, the analogue is an affordable-action envelope or score-deficit cost derived from the agents' cost geometry. In fixed-charge and capped models, pressure creates active-set switches and lumpy spillover. These are not the same prediction, which is the point. Once the response geometry is declared, different mathematical tools predict different distortion shapes.

The multidimensional scorecard results are also promising because they explain why proxy repair can have signs that scalar Goodhart cannot express. Under an additive compensatory score, adding an independently gameable dimension can lower the cheapest cost of clearing a score deficit and recruit more agents into gaming, raising H_{pop} through entry. Under a conjunctive score, the same added dimension can block substitution but raise per-gamer burden and false-negative risk, so the population sign depends on entry. Conservation of H_{per} appears only under an exchange-rate condition: hidden harm must be proportional to score contribution on the active channels. This is the kind of claim I want from the paper: not "more metrics are bad", but "given this aggregation rule and these exchange rates, here is the distortion geometry". Notice the parallel with the selection case, but also the difference. For selection, the basic question is how the added metric depends on hidden coordinates in the baseline distribution. For intervention, the basic question is action geometry: which score-improving moves the metric opens, blocks, or makes redundant, and how those moves affect hidden value. The two can be related empirically, but they are not the same mathematical primitive.

The live work-in-progress questions (collected as a bulleted inventory below) belong here in the spine rather than as a separate shopping list. A general convex intervention bound is promising if it stays conditional on action geometry, stochastic response, stakes, and hidden harm. Application evidence standards are promising if they force each case to declare type, response kernel, action traces, and falsifiers. Adaptive hardening is promising as a dynamic question: hardening the currently exploited channel lowers κ_j, but whether that converges faster than choosing a narrow hard metric from the start is open. The measurement frontier is promising because it names the regulator's real tradeoff: measure enough dimensions to recover signal while not adding so much independent gaming capacity that the attack surface wins.

1.4 What does not look promising

Several tempting framings should be rejected by name.

Unconditional dimensional scaling does not work. More hidden coordinates do not automatically imply more Goodhart; the claim needs a coupling model and a harm functional.

Baseline covariance as a universal primitive does not work. With nonlinear tail dependence, hidden drift can be zero-covariance and still move under threshold or finite soft selection.

Absolute continuity as the causal intervention boundary does not work. If the baseline has even epsilon mass on gaming-like behavior, an announced metric can change fixed-type behavior while the post-policy law remains absolutely continuous. The boundary has to be response-kernel change relative to a declared type representation.

Generic minimum-complexity attraction does not work. Quadratic costs select minimum-cost directions, which can be dense. Selection follows baseline tails. Fixed charges, low-rank affordances, or search priors can produce sparse or low-description-length failures only when those mechanisms are specified first.

Strong additive conservation does not work. Re-routing conserves $H_{\mathrm{per}(M,d)}$ only when harm-per-score exchange rates match. Change weights, harm coefficients, or the participating population, and the relevant welfare object changes.

The scalar-κ mapping to neural training does not work yet. In the toy model, κ is ease of gaming. In RLHF or benchmark-driven finetuning, the analogue might be gradient accessibility, pretraining density, benchmark contamination, reward-model feature simplicity, or optimizer search efficiency. Those are not interchangeable.

So the rest of the paper answers a narrower and better question: does multidimensional Goodhart make proxy-induced distortion modelable, by predicting the shape of distortion from the declared response channel, aggregation rule, and action geometry, in regimes where scalar Goodhart can only say that optimization pressure broke the proxy?

1.5 Work in progress

Threads that are live in the sense that fragments of math exist but are not yet promoted into the formal chapters. They shape what the rest of the book is trying to prove.

- **Convex intervention budget.** The selection-channel bound governs hidden drift through a chi-square / reweighting budget on baseline statistics. The intervention analogue (Section 4.6) gives a score-deficit cost after declaring a finite-dimensional action space, linear score gain, convex agent cost, and regularity condition. The quadratic case recovers the $\sqrt{2\kappa V}$ wedge. What remains open is not the convex-duality calculation, but the application mapping: stochastic response, participation, hidden harm, and the link between selection value and realised action cost.

- **Evidence standards / domain templates.** Chapter 5 states the response-modeling contract abstractly. The in-progress work is per-domain instantiation — ML evaluations, RLHF and finetuning, institutional scorecards, scientific metrics — each naming concrete referents for κ, V, type space U, response kernel K_θ, action/cost geometry, aggregation rule, and hidden welfare model. The template must force a declaration of falsifiers and action traces, not degenerate into a generic checklist.
- **Adaptive hardening.** A regulator that each period lowers κ_j on whichever measured channel is currently most-gamed drives aggregate gaming capacity K_M downward. Open: does this policy converge to no-gaming ($K_M < t^2/(2V)$), and at what rate? Does it dominate the alternative of committing from the start to a narrow hard-to-game measured set M?
- **Measurement frontier.** The dual of hardening. More measured channels recover more real signal about G but enlarge the gaming attack surface K_M. Characterise the frontier between "informative enough" and "small enough attack surface" — including whether the frontier is sometimes empty (no good choice exists).
- **Response-shape simulation comparison.** The Chapter 4 conditional menu — quadratic costs spread drift along Cw; fixed activation or linear costs concentrate drift in the uncapped no-tie case; caps and convex post-activation costs spread it again, sometimes lumpily; low-rank action maps restrict drift to $\mathrm{im}(L)$ — has not been simulated head-to-head under a fixed hidden coordinate system and a pre-specified shape measure (support size, rank, KL from a max-entropy prior, description length under a fixed coding). This is the cheapest next test of the menu before any row is upgraded.
- **Recursive toy model.** Appendix E draws a cartoon; the formal chapters do not prove that successive proxy patches push residual error into less-legible dimensions. Needed: a two- or three-round refinement model with declared hidden dimensions, a declared response geometry, and a legibility measure fixed **before** the patching sequence is used as evidence. Without that, the recursive hypothesis is not yet testable.
- **Per-agent versus population welfare.** The additive gaming section now separates fixed-deficit harm $H_{\mathrm{per}(M,d)}$ from population harm $H_{\mathrm{pop}}(M, F_Q, V)$. In the unit-weight equal-harm model, $H_{\mathrm{pop}} = \mathbb{E}\left[(t-Q)\mathbb{1}\left\{0 < t - Q \le \sqrt{2K_M V}\right\}\right]$ for heterogeneous quality Q. Open: which population welfare object is right under noise, nonlinear costs, and endogenous V?
- **ML mappings for $\boldsymbol{\kappa}$.** In the toy Stackelberg model κ is scalar ease of gaming. In RLHF, finetuning, and benchmark-driven optimisation, candidate referents include gradient accessibility, pretraining density, benchmark contamination, reward-model feature simplicity, and optimizer search efficiency. These are not interchangeable. Open: which carry the load of κ, jointly or separately, and which (if any) are measurable from training-time signals rather than only post-hoc.

- **Endogenous stakes.** Selection value V is treated as exogenous, but trust in a metric decays as gaming is observed, so V falls as the metric is used. Open: does the regulator–agent game with endogenous V have a performative-stable fixed point, and does that fixed point still carry hidden harm rather than collapsing to zero gaming and zero stakes? A tempting toy closure is $V(H_{\mathrm{obs}}) = V_0 \exp(-\gamma H_{\mathrm{obs}})$, but this book does not solve that fixed point. Endogenous V changes the response channel itself and needs a separate performative-response model specifying what H_{obs} is, how agents forecast it, and how the regulator updates the metric. Until then, all Stackelberg and convex-cost calculations use exogenous V.

1.6 Open questions

A separate class of questions surfaced during this work and is deliberately **not** on the critical path. Each needs a substantial new piece of machinery; none of them blocks Chapters 1–5. They are recorded here so they are not rediscovered from scratch.

- **Spectral / basis decomposition of the error.** Three decompositions carrying different information should be kept distinct: the SVD of φ (right singular vectors are the G-directions φ distinguishes; small singular values mark structural blind spots in the **dimension gap**); the eigendecomposition of $\mathrm{Cov}(\varepsilon)$ (principal directions of measurement noise, the **observation gap**); and the selection-induced shift of G before vs after restricting to the selected set — the most diagnostic of actual Goodhart and the least obvious to compute. Bad-case corner: a G-direction where φ has small singular value, $\mathrm{Cov}(\varepsilon)$ has mass, and the selection shift is large. Each decomposition is easy individually; open whether the joint object is useful without first declaring a value-weighting on G-space.
- **Value-weighted Goodhart susceptibility.** PCA/SVD is value-neutral; Goodhart is value-laden. A huge principal component orthogonal to the value structure of G is harmless; a tiny one aligned with a load-bearing G-coordinate can dominate. Proposed scalar per G-direction: (value importance) $\times$ (1/ singular value of φ) $\times$ (noise variance) $\times$ (selection shift), under a value-weighted inner product on G-space. Open: whether reasonable proxies for value importance (the principal's true utility gradient or a proxy for it) recover useful structure, and whether the value-weighted version closes the geometric blind spot of PCA-based detection against value-aware adversaries who route ε into low-variance components of historical observation.
- **Pareto selection rather than scalarisation.** Chapter 2 fixes a weighted score and then a threshold. Pareto-frontier selection is a different operator: it preserves diversity across measured dimensions in a way scalar thresholding destroys. Open: when does selecting the Pareto-undominated set compress the selected distribution enough to break identifiability of the **unmeasured** dimensions, and when does it not? Chapter 2's selection-channel response theory does not transport to it for free.
- **φ-as-partial-observation alternative.** The chapters fix $\varphi : \mathbb{R}^m \to \mathbb{R}^k$ as the principal's intended correspondence and read $\ker \varphi$ as the dimension gap. An

alternative represents the proxy as a partial observation of G plus independent artifacts; this changes what "$\ker \varphi$" means and may interact better with partial-hypothesis formalisms in the style of infra-Bayesianism. Worth a side-by-side comparison once a live thread settles; doing it prematurely risks rebuilding the same machinery in a different alphabet.

- **Incomputable noise / over-dimensional agents.** If a human writes a proxy intended to align a system whose objectives have **more** dimensions than the human's, the proxy contains error in directions the human's G has no coordinate for — error that cannot be scored, only suffered. Open: whether partial hypotheses (infra-Bayesian style) recover any handle on this, and whether there is any reason to expect a more capable system to have **fewer** goal dimensions than its principal rather than more. Recorded because the multidimensional framing makes the question articulable; parked because none of the current tools answers it.

The shared property of the five parked questions is that each requires new machinery — a value-weighted norm on G-space, a Pareto-selection response operator, a partial-observation rephrasing of φ, a partial-hypothesis formalism, or a precise sense in which the agent has more goal dimensions than the principal. They are blocked on the right object to define, not on effort.

2 Goodhart’s law, multidimensionally

2.1 The shape of the problem

Goodhart’s law is usually quoted in one of two forms. The popular slogan, which is actually Strathern’s compression [15]:

> When a measure becomes a target, it ceases to be a good measure.

and Goodhart’s own statement [6]:

> Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes.

Both are stated about a **scalar** measure: a single number that was correlated with something we cared about, until we started optimising it. The thesis of this book is that the scalar framing hides most of the structure. Real goals — the goals of a person, a team, a company, a state, or a trained model — are **multidimensional**, and the proxies we use to steer toward them are multidimensional too, usually with a different (and smaller) set of dimensions. Once that is taken seriously, “the regularity collapses” is replaced by a more precise and more useful question: **how does pressure on a proxy reshape the distribution of its residual error?**

This is not a rejection of scalar analysis. Sometimes the principal really has a declared scalar objective, or the application has a defensible value-weighted loss. Then scalar Goodhart is the right projection. The problem appears when the scalar is introduced only because the dashboard, benchmark, or theorem wants one number. A vector residual has length and direction only after a norm or value metric is chosen. Without that choice, saying that “proxy error increased” is ambiguous; with the wrong choice, the model can hide exactly the direction in which optimisation caused damage. Treating each coordinate as a separate scalar problem avoids one compression but loses the cross-coordinate geometry: substitution, cancellation, bottlenecks, active constraints, and tail dependence. Those cross-terms are where many proxy repairs succeed or fail.

The answer depends on the channel. In a **selection** regime, the policy reweights a fixed baseline distribution; hidden drift is governed by the baseline response of hidden coordinates to the selected proxy, and dimensionality matters only through coupling and variance structure. In an **intervention** regime, agents respond; the response kernel changes at a fixed underlying type, so mass can move to states the old baseline assigned little or no probability. The drift is governed by cost geometry, stakes, and available gaming channels. A broader recursive-Goodhart hypothesis is that repeated proxy refinement often pushes the remaining error into dimensions that are less legible to the evaluator. This book treats that as a hypothesis suggested by the framework, not as a theorem of the framework.

So there are three layers to keep separate. First, the formal chapters prove or derive toy results about selection response, baseline drift bounds, intervention channels, multidimensional gaming, and response-modeling contracts. Second, those models imply conditional design lessons: additive scorecards, conjunctive scorecards, and different harm-per-score exchange rates behave differently. Third, the recursive story — that residual error may migrate into less-monitored or harder-to-elicit dimensions as the proxy stack is patched — is an empirical conjecture. The job of these first chapters is to make all three layers statable without pretending they have the same epistemic status.

> *Remark.* **Licensed claims.** These chapters license five narrow uses. First, when pressure only reweights a fixed baseline distribution, the right quantities are selection-response functionals of that baseline: covariance locally, threshold response in the tails, and a chi-square drift budget for bounded selection. Second, when agents can change the state-generating process, the drift budget is not a baseline statistic; in the quadratic Stackelberg toy model it is $\sqrt{2\kappa V}$, a function of gaming ease and stakes. Third, for intervention channels with compensatory additive scores and separable quadratic gaming costs, adding an independently gameable measured dimension increases aggregate gaming capacity K_M, lowers the minimum cost of clearing a fixed score gap, and weakly expands the population of agents for whom gaming pays; this is a population-harm statement, while fixed-deficit H_{per} is conserved only under the exchange-rate condition. These claims are meant for designing scorecards, benchmarks, and evaluation suites. Fourth, proxy pressure does not generically select a minimum-complexity residual; a response-shape prediction needs a named response geometry, constraint set, and complexity or shape measure. Fifth, a Goodhart claim is incomplete until it declares the response model: type, action, cost, aggregation, hidden welfare, and evidence standard. These claims do not prove the recursive-Goodhart hypothesis, do not say that "more metrics is worse" as a rule, and do not apply to RLHF or other non-convex training dynamics until the application has declared a response model that matches their assumptions.

The starting point is the variant taxonomy of Manheim and Garrabrant [10] — regressional, extremal, causal, and adversarial Goodhart — and Wentworth's geometric reconstruction of the regressional case [16], plus Wentworth's later warning that experiments and metrics often measure a different latent quantity than their designers think they do [17]. The economic ancestor is the multitask principal-agent line: Holmstrom and Milgrom's canonical model of measurable and unmeasurable tasks [9], Baker's performance-measurement model [1], and Prendergast's survey of incentive provision [13]. In AI safety, the closest formal neighbours are reward gaming and misspecification models [11, 14] and the partial-objective model of misaligned AI [18]. The contribution here is to put these literatures in one frame: the Goodhart taxonomy supplies the failure modes, strategic classification and performative prediction supply adaptive distribution shift, principal-agent theory supplies substitution toward measured tasks, and the selection/intervention split makes the vector structure of both regimes explicit.

2.2 A worked intuition: the hierarchy of proxies

Consider a company with leadership whose real objectives are something like *make a product customers want*, *make money*, *enjoy the work*, *gain prestige*, and a dozen other things, in some weighting that no written document captures. Call that vector of objectives G. Each quarter the leadership sets a proxy P: a handful of metrics — revenue, delivery dates, an NPS number. The teams that implement P have their own objectives, and the proxy they **actually** optimise, P_i, folds their incentives into the official one. Its dimensionality is at least that of the most complex implementer's goals.

Now suppose leadership notices that Team 1 is "optimising too hard on the metrics" — the dashboard looks better than the product. They intervene. The correction they apply is *not* the gap between G and Team 1′s true output. It is the gap between the **felt proxy** by which leadership noticed the problem and Team 1′s measured performance. The error term is being filtered through several layers of proxy, each with its own residual.

Two things follow, and they recur throughout the book:

1. When a principal constrains an agent whose behaviour has more degrees of freedom than the principal's proxy, the proxy must compress, ignore, or invent coordinates. Some residuals are ordinary measurement errors. Others are closer to incomputable noise: not noise the principal is failing to measure well, but behaviour whose relation to the principal's goal has not been represented in the proxy space at all. This is the **dimension gap**: if the proxy map has no coordinate for a goal-relevant degree of freedom, optimisation can move that degree of freedom without registering as proxy error. The later formal claims are about special cases where that movement is observable as either selection response or intervention cost.

2. Whether the multi-layer correction process converges depends on whether the layers' perception errors are roughly independent. If each layer's residual is its own, a dimension of error that one layer sees can be corrected even though no layer sees the whole picture. If the residuals are strongly correlated — a centrally-planned organisation, a single dominant metric — the system is fragile: every layer is blind in the same direction at once.

Remark. There is an ML reading of all this, but it is diagnostic rather than a finished model of training. Pretraining, validation-set selection, and model selection are selection regimes when they choose among behaviours already present in the candidate distribution. RLHF and reward-model optimisation become intervention-like once the learner changes its policy to exploit the reward channel; this is the reward-gaming setting studied by [14] and [11]. What plays the role of κ in a neural network is not settled: it might be gradient accessibility, representation density from pretraining, benchmark contamination, reward-model feature simplicity, or optimiser search efficiency. The framework below therefore does not claim that quadratic Stackelberg gaming describes RLHF. It says what an

ML version would have to estimate: baseline response curves for selection; cost geometry, stakes, and available gaming channels for intervention. If those objects have not been declared, the application is underspecified; that is not evidence for or against the framework's toy bounds.

2.3 Setup: goals, proxies, and two kinds of gap

Fix a state space S — the set of configurations the world (or the organisation, or the model) can be in. A **true goal map** is a function $G : S \to \mathbb{R}^m$; its m coordinates are substantively distinct things the principal cares about. A **proxy map** is $P : S \to \mathbb{R}^k$, the k things actually measured and acted on. The principal also has in mind an **intended model** $\varphi : \mathbb{R}^m \to \mathbb{R}^k$ — "the proxy should be roughly φ of the goal" — and the discrepancy is the **residual**

$$\varepsilon(s) = P(s) - \varphi(G(s)).$$

This little decomposition already separates two distinct Goodhart channels:

- A **dimension gap**: $\ker \varphi \neq \{0\}$, so some directions in goal space are invisible to the proxy. No amount of measuring P accurately tells you anything about movement along $\ker \varphi$.
- An **observation gap**: $\varepsilon \neq 0$, so even the directions the proxy is meant to track are corrupted — by noise, by lag, or (Chapter 3) by deliberate manipulation.

> **Toy example.** Let product quality be $G = (\text{reliability}, \text{delight})$ and let the proxy be uptime. Then "delight" lies in $\ker \varphi$ — the dimension gap. Noisy uptime instrumentation is part of ε — the observation gap. A research-evaluation example: citation count is **intended** to track research quality; "long-run conceptual fertility" may be in $\ker \varphi$, while bot citations and citation cartels are residual variation in proxy space.

Remark. A caution that will shape the notation. If the principal's "true" objective is genuinely scalar, then $\ker \varphi \neq \{0\}$ may be an artefact of having written that scalar goal in redundant coordinates. So dimension-gap claims should always be made **after** choosing coordinates whose components are substantively distinct goal variations, not arbitrary embeddings. We will keep $\varphi : \mathbb{R}^m \to \mathbb{R}^k$ as the principal's intended correspondence and consistently distinguish three subspaces:

- $\ker \varphi$: goal variation invisible to the intended proxy;
- $\operatorname{im} \varphi$: proxy variation intended to track goal variation;
- residual proxy directions: variation of P not explained by $\varphi(G)$.

2.4 A first specialisation, and a roadmap

Throughout Chapter 2 we work the simplest non-trivial case: $G(s) = X \in \mathbb{R}^m$ with X Gaussian, a single scalar proxy that is one coordinate (or one linear combination) of X plus noise, and **selection** — keep the states scoring above a threshold. This isolates the two classical "easy" Goodhart effects, **regressional** and **extremal** Goodhart [10], from the harder ones. It is the right first probe precisely because it is the regime where, as Chapter 2 shows, hidden harm turns out to be **bounded by quantities visible in the pre-selection distribution**. That selection analysis stops at reweighting: it does not say what fixed agents can do after a metric is announced. Chapter 3 therefore starts again from declared response kernels, actions, costs, stakes, and aggregation geometry. It shows where the Chapter 2 baseline-only bound no longer has an analogue, and what must replace it.

A reader who wants the punchline before the construction: there is a clean dichotomy between **selection channels** (the policy reweights a fixed baseline distribution; all classical regressional/extremal Goodhart lives here; hidden drift $\leq \delta \cdot \|s\|$ with every term a baseline functional) and **intervention channels** (the policy moves probability mass to where the baseline had none; causal and adversarial Goodhart live here; no baseline bound exists, and any bound must be imported from a model of what the responding agents can afford to do). Chapters 2 and 3 are those two halves.

The appendices are organised the same way. Appendices A–D are visual aids for claims made in the formal chapters: selection response, dimensional coupling, selection versus intervention, additive versus conjunctive gaming, and the exchange-rate condition for conservation. Appendix E is deliberately different: it is a speculative cartoon of the recursive-Goodhart intuition. Its purpose is to say what the framework might help test, not to smuggle an additional theorem into the paper.

2.5 Summary of results

The proved or directly derived results in these chapters are:

- Gaussian threshold selection shifts hidden means by $\mathbb{E}[H \mid A_t] = \sigma_1 \lambda(t/\sigma_1) r$ in the linear-Gaussian model.
- Covariance is not a universal coupling primitive: with $P = Z$ and $H = Z^2 - 1$, baseline covariance vanishes while threshold and finite Boltzmann selection still move H.
- Under Boltzmann selection on the finite-mgf domain $\mathcal{B} = \left\{\beta : \mathbb{E}_{\mu[\exp(\beta P)]} < \infty\right\}$, covariance is the local velocity at differentiability points: $\mathrm{d}\mathbb{E}_{\beta[H]} / \mathrm{d}\beta = \mathrm{Cov}_{\beta(H,P)}$.
- Selection-channel drift satisfies the coordinate-explicit bound $\left\|B_{H(\theta)}\right\|_2 \leq \delta \cdot \|s\|_2$, with $\delta^2 = \chi^2(\mu_\theta \parallel \mu_0)$.
- The same selection argument gives the declared value/operator bound $|\Delta(v \cdot H)| \leq \delta\sqrt{v^T \Sigma_H v}$, and its dual-norm form after a value metric is declared.
- In the quadratic Stackelberg gaming model, the gaming wedge is $\Delta = \sqrt{2\kappa V}$.
- In the additive multidimensional gaming model, quadratic costs give the water-filling allocation, and the weighted additive case gives the exchange-rate condition $h_j = cw_j$ for conservation of fixed-deficit $H_{\mathrm{per}(M,d)}$.
- In the quadratic response-shape model, an intervention target $w \cdot a \geq d$ with cost $\left(\frac{1}{2}\right) a^T C^{-1} a$ selects the unconstrained/interior minimum-cost direction $a^* = dCw/(w^T C w)$, not a minimum-complexity direction as such; sign constraints require active-face solutions.
- In fixed-charge or linear-cost response models, the uncapped no-tie case can produce one-channel drift; caps convert this into ordered spillover only when

activation costs are absent or already paid; positive activation costs add thresholded regime switches.

The additive-versus-conjunctive flip and the noisy Stackelberg refinement are illustrative models, not general theorems. The convex-cost intervention analogue of the selection bound is stated below as a score-deficit proposition with a Fenchel-duality proof sketch. Chapter 4′s response-shape taxonomy is a conditional prediction menu, not a theorem that residual error generically becomes more complex. The live open questions are pulled forward into the intro as modelling obligations.

3 Selection channels: when the principal only re-selects

3.1 The Gaussian threshold model

Let $X = (X_1, ..., X_m) \sim \mathcal{N}(0, \Sigma)$. Take the proxy to be the first coordinate, $P = X_1$, with selection $A_t = \{X_1 \geq t\}$; the **hidden** goal coordinates are $H = (X_2, ..., X_m)$, so in this stripped-down picture $\dim(\ker \varphi) = d = m - 1$. (We have set the observation noise to zero on purpose: whatever effect appears here comes from the dimension gap, not from measurement error.)

For each hidden coordinate write $\rho_j = \frac{\mathrm{Cov}(X_j, X_1)}{\mathrm{Var}(X_1)}$. The Gaussian conditional-mean formula gives $\mathbb{E}\left[X_j \mid X_1 = x\right] = \rho_j x$, and hence

$$\mathbb{E}\left[X_j \mid A_t\right] = \rho_j \cdot \mathbb{E}[X_1 \mid X_1 \geq t].$$

If $X_1 \sim \mathcal{N}(0, \sigma_1^2)$ then $\mathbb{E}[X_1 \mid X_1 \geq t] = \sigma_1 \lambda(\alpha)$ with $\alpha = \frac{t}{\sigma_1}$ and $\lambda(\alpha) = \frac{\phi(\alpha)}{1 - \Phi(\alpha)}$ the inverse Mills ratio. So the hidden mean-shift vector is

$$\mathbb{E}[H \mid A_t] = \sigma_1 \lambda(\alpha) \cdot r, \qquad r = (\rho_2, ..., \rho_m).$$

> **Selection moves only what correlates with the proxy (in this model).** In the Gaussian threshold model, thresholding on X_1 produces no expected drift in a hidden coordinate X_j exactly when $\mathrm{Cov}(X_j, X_1) = 0$. *Toy example:* if a school rewards only test scores and student curiosity is statistically independent of test scores in the applicant pool, thresholding on scores does not change mean curiosity.

Remark. This is a **Gaussian** fact and must not be read as a general one — see Section 3.3. It is true here only because Gaussian conditional expectations are linear; for non-Gaussian variables, zero covariance does not imply a flat tail-conditional mean. Read correctly, it is a **sufficiency** statement: in the multivariate-Gaussian scalar-threshold model, the covariance ratios r are a complete summary of hidden mean drift at every threshold.

3.2 Does the harm scale with the number of hidden dimensions?

The motivating intuition wants "more hidden dimensions ⇒ more Goodhart". The honest version is weaker, and Chapter 2's final bound gives the clean statement. This Gaussian calculation is only the first view.

If every hidden dimension has the **same** covariance ratio $\rho_j = \rho$, then

$$\|\mathbb{E}[H \mid A_t]\|_2 = \sqrt{d} \cdot |\rho| \cdot \sigma_1 \lambda(\alpha),$$

so the aggregate hidden mean drift grows like $\sqrt{\dim(\ker \varphi)}$ at fixed per-dimension coupling. But "fixed per-dimension coupling" is a strong scaling assumption. If instead the **total** correlation budget is bounded, $\sum_j \rho_j^2 \le C$, then $\|\mathbb{E}[H \mid A_t]\|_2 \le \sigma_1 \lambda(\alpha)\sqrt{C}$, which does not grow with d at all. Whether dimensional growth is real depends entirely on whether new dimensions bring new **independent** coupling to the proxy or merely subdivide a fixed amount of coupling into finer coordinates.

> The clean first version of dimensional dependence is therefore **not** "more hidden dimensions automatically cause more Goodhart". It is: under a per-dimension coupling model, threshold selection induces hidden drift whose **norm scales with the coupling-vector norm** $\|r\|_2$; this becomes **dimensional** scaling only after a substantive assumption about how $\|r\|_2$ grows with d.

This is the same lesson that the selection-channel drift bound will express in a declared hidden coordinate system: dimension enters through the hidden variability and coupling budget, not through a bare count of unmeasured coordinates. Appendix A visualises both parts of the claim.

Remark. Two shortcuts fail. If hidden dimensions are independent of the proxy, thresholding leaves them unchanged. And the signed aggregate hidden error is the wrong target: positive and negative hidden correlations can cancel while normed or squared displacement grows.

3.3 Covariance is not a general coupling primitive

Outside Gaussian linearity, covariance fails. Let $P = Z$ with $Z \sim \mathcal{N}(0,1)$ and let the hidden coordinate be

$$H = Z^2 - 1.$$

Then $\mathbb{E}[H] = 0$ and $\mathrm{Cov}(H, P) = \mathbb{E}[(Z^2-1)Z] = \mathbb{E}[Z^3] - \mathbb{E}[Z] = 0$. But under threshold selection $A_t = \{Z \ge t\}$,

$$\mathbb{E}[H \mid A_t] = \mathbb{E}\big[Z^2 - 1 \;\big|\; Z \ge t\big] = t\lambda(t) > 0 \quad \text{for } t > 0.$$

> Zero covariance between a hidden coordinate and a scalar proxy does **not** imply zero hidden drift under threshold selection. *Toy example:* an academic evaluation score may be uncorrelated with intellectual conformity overall — because both very-low-score and very-high-score candidates are unusual — while selecting only the very highest scores still enriches for one particular kind of unusualness.

Remark. The example writes H as a deterministic function of P, which feels engineered; but that is exactly the point against covariance — nonlinear dependence can be invisible to covariance while completely determining tail behaviour. A less deterministic version $H = Z^2 - 1 + \xi$ with independent mean-zero ξ keeps the same covariance and the same tail-mean shift while weakening the functional-dependence objection. The right takeaway is **not** that real systems generically have U-shaped hidden dependence; it is that covariance cannot be the universal coupling primitive.

The repair is to take the **response curve itself** as primitive. For a scalar proxy P and hidden vector H, define the **threshold response**

$$b_{H(t)} = \mathbb{E}[H \mid P \geq t] - \mathbb{E}[H],$$

whenever the conditional expectation exists. This describes the actual displacement caused by selection at pressure level t, with no linearity assumption. In the Gaussian scalar model it reduces to what we already have: $b_{H(t)} = \sigma_1 \lambda\left(\frac{t}{\sigma_1}\right) r$, so r is a sufficient statistic for **all** threshold responses in that restricted model — but only there. See Figure 1 for the corresponding three-case picture.

> *Remark.* $b_{H(t)}$ is a **threshold-selection** primitive, not the final one. It says nothing about smoother optimisation policies — funding everyone above a cutoff is one thing; funding probabilistically with odds increasing in score is another. The next section generalises.

3.4 Weighted selection response

Let $(S, \mathcal{F}, \mu)$ be the baseline probability space, $H : S \to \mathbb{R}^d$ the hidden coordinates, $P : S \to \mathbb{R}$ the scalar proxy. A **selection policy** is a nonnegative weight $W_\theta : S \to [0, \infty)$ with $0 < \mathbb{E}_{\mu[W_\theta]} < \infty$. Define the **selected expectation** and the **hidden response**

$$\mathbb{E}_{\theta[F]} = \frac{\mathbb{E}_{\mu[F W_\theta]}}{\mathbb{E}_{\mu[W_\theta]}}, \qquad B_{H(\theta)} = \mathbb{E}_{\theta[H]} - \mathbb{E}_{\mu[H]}.$$

Hard thresholding is the special case $W_t = \mathbb{1}\{P \geq t\}$ (recovering $b_{H(t)}$). Soft optimisation by Boltzmann weights is available only on the finite-mgf domain

$$\mathcal{B} = \left\{\beta \in \mathbb{R} : \mathbb{E}_{\mu[\exp(\beta P)]} < \infty\right\}.$$

For $\beta \in \mathcal{B}$, write $W_\beta = \exp(\beta P)$. Probabilistic funding with score-increasing odds, replicator-style repeated reweighting by performance, top-q-fraction selection (thresholding at an endogenous quantile): all are weight functions when their normalizing expectations are finite.

> *Remark.* Two caveats on the soft-optimisation case. First, if P is heavy-tailed, $\mathbb{E}[\exp(\beta P)]$ may be infinite for positive β — the Boltzmann path may not exist. This is not a technicality; Goodhart is often precisely about extreme tails. For heavy tails, bounded weights or quantile selection are safer models. Second, not every control process is a reweighting of a fixed baseline at all — interventions can change the state-generating mechanism. That is Chapter 3.

3.4.1 Covariance as a local velocity

For Boltzmann selection on the finite-mgf domain $\mathcal{B}$, $\mathbb{E}_{\beta[H]} = \frac{\mathbb{E}[H \exp(\beta P)]}{\mathbb{E}[\exp(\beta P)]}$. At values of β where the tilted expectation of HP is finite and differentiation through the normalizer is valid — in particular at interior points of a finite neighborhood inside $\mathcal{B}$ under the usual domination conditions —

$$\frac{\mathrm{d}}{\mathrm{d}\beta} \mathbb{E}_{\beta[H]} = \mathbb{E}_{\beta[HP]} - \mathbb{E}_{\beta[H]} \mathbb{E}_{\beta[P]} = \mathrm{Cov}_{\beta(H,P)}.$$

Covariance is best read as the **local velocity** of hidden drift under infinitesimal soft optimisation only where the Boltzmann path exists and is differentiable, evaluated under the **current** selected distribution — not as a global finite-pressure summary. *Toy example:* at low bonus pressure, the rate at which burnout changes with sales incentives is the current covariance between burnout and sales; after employees adapt or the selected population shifts, the covariance must be recomputed under the new weighted distribution.

That local velocity does not pin down finite movement. Take again $P = Z \sim \mathcal{N}(0,1)$, $H = Z^2 - 1$. Boltzmann tilting by $\exp(\beta Z)$ gives $Z_\beta \sim \mathcal{N}(\beta, 1)$, so

$$\mathbb{E}_{\beta[H]} = \mathbb{E}\left[Z_\beta^2 - 1\right] = \beta^2.$$

The covariance at $\beta = 0$ is zero — matching Section 3.3 — yet finite pressure gives strictly positive hidden drift for every $\beta \neq 0$. Baseline covariance alone is insufficient even for **finite** soft optimisation; one needs the covariance field $\mathrm{Cov}_{\beta(H,P)}$ along the whole tilted path, and the path itself must remain inside $\mathcal{B}$. Heavy-tailed proxies for which $\mathbb{E}[\exp(\beta P)] = \infty$ at the pressure of interest are not covered by the Boltzmann calculation; they require bounded, truncated, or quantile-based selection models.

The hierarchy of selection primitives. covariance (infinitesimal Boltzmann velocity) $\subset$ threshold response $b_{H(t)}$ (hard cutoffs) $\subset$ weighted response $B_{H(\theta)}$ (generic non-causal selection). All three are functionals of the baseline μ alone. Causal and adversarial Goodhart need a further layer in which μ itself changes with the principal's policy — the subject of Chapter 3.

3.5 The selection-channel drift bound

This is the structural payoff of staying inside the reweighting picture. With W_θ a selection policy, write the likelihood ratio $L_\theta = \frac{W_\theta}{\mathbb{E}_{\mu[W_\theta]}}$, so the selected law is $\mu_\theta = L_\theta \mu$. By Cauchy–Schwarz in $L^2(\mu)$, for each hidden coordinate H_i,

$$\left|B_{H_i}(\theta)\right| = \left|\mathbb{E}_{\mu\left[(L_\theta - 1)\left(H_i - \mathbb{E}_\mu H_i\right)\right]}\right| \leq \|L_\theta - 1\|_{L^2(\mu)} \cdot \left\|H_i - \mathbb{E}_\mu H_i\right\|_{L^2(\mu)}.$$

So, writing $\delta := \|L_\theta - 1\|_{L^2(\mu)}$ — a **reweighting budget**, with $\delta^2 = \chi^2(\mu_\theta \parallel \mu)$ the chi-square divergence — and $s_i := \mathrm{sd}_{\mu(H_i)}$,

$$\left|B_{H_i}(\theta)\right| \leq \delta \cdot s_i, \qquad \left\|B_{H(\theta)}\right\|_2 \leq \delta \cdot \|s\|_2.$$

Proposition 1 (coordinate-explicit selection drift). Under a pure selection channel with likelihood ratio $L_\theta = d\mu_\theta / d\mu$, finite coordinate variances, and $\delta = \|L_\theta - 1\|_{L^2(\mu)} = \sqrt{\chi^2(\mu_\theta \parallel \mu)}$, each hidden coordinate satisfies

$$\left|B_{H_i}(\theta)\right| \leq \delta s_i, \quad \left\|B_{H(\theta)}\right\|_2 \leq \delta \|s\|_2.$$

> Licenses: after the hidden coordinates and Euclidean norm are declared, selection-regime drift is bounded by reweighting intensity and baseline hidden variability. The $\sqrt{d}$ growth from Chapter 2′s per-dimension model is the $\|s\|_2$ term.
>
> Does not license: a coordinate-free welfare claim, or a value metric inferred from the selected law.

For a declared scalar hidden-value functional $v \cdot H$, the same proof gives

$$|\Delta(v \cdot H)| \leq \delta\sqrt{v^T \Sigma_H v},$$

where $\Sigma_H = \mathrm{Cov}_{\mu(H)}$. More generally, for any hidden-drift value norm $\|.\|_V$,

$$\left\|B_{H(\theta)}\right\|_V \leq \delta \sup_{\|v\|_{V,*} \leq 1} \sqrt{v^T \Sigma_H v}.$$

If $\|x\|_M = \sqrt{x^T M x}$ for a positive definite value matrix M, this is the corresponding covariance-operator bound after the value metric has been declared.

> **Proposition 1' (value-weighted/operator selection drift).** Let $L = d\mu_\theta / d\mu$, $L \in L^2(\mu)$, $\mathbb{E}_{\mu[L]} = 1$, and let H have finite second moments. For every declared value vector v,
>
> $$\left|\mathbb{E}_{\mu_\theta}[v \cdot H] - \mathbb{E}_{\mu[v \cdot H]}\right| \leq \sqrt{\chi^2(\mu_\theta \parallel \mu)}\sqrt{v^T \Sigma_H v}.$$
>
> The dual-norm/operator form above is the same statement optimized over the declared value unit ball.
>
> Licenses: coordinate bookkeeping is repaired once the application supplies a scalar value direction or value norm. Splitting a hidden coordinate does not change the bound for the same underlying value functional.
>
> Does not license: hidden value weights observable from μ_θ, or a useful numerical prediction when χ^2 is huge or infinite. In that case the inequality remains valid where defined, but becomes practically vacuous.

Remark. The proof is only Cauchy–Schwarz, but the formulation is doing real work. It is the clean coordinate-explicit statement of why selection-regime Goodhart is tame after the hidden coordinates and norm have been chosen: if a policy only reweights a baseline distribution, then hidden drift is bounded by a reweighting budget and baseline hidden variability, both μ-functionals. It is not invariant to arbitrary relabellings of the hidden space. Splitting one hidden variable into ten correlated coordinates changes the bookkeeping unless the norm is replaced by a declared covariance or value-weighted operator norm. The caveats are still important: this is a worst-case envelope, not a prediction, and it can be vacuous under extreme selection when δ is large. The rest of the book is about what breaks when the policy is not a reweighting and no such baseline-only budget exists.

4 Intervention channels: when agents respond

4.1 Why selection is not enough

A selection policy can only move probability mass that already exists toward high-P regions: the selected law $\mu_\theta = L_\theta \mu$ is absolutely continuous with respect to μ, so any event with baseline probability zero keeps probability zero. That measure-theoretic criterion is a useful sufficient formalism for pure selection, but it is not the causal boundary by itself. If the baseline already assigns tiny probability to test-specific drilling or fabrication, an announced metric can induce much more of the same behaviour while the new law remains absolutely continuous. The causal distinction is that the policy changes the response kernel at a fixed underlying type: agents can move their measured and hidden features, not merely be reweighted.

Teaching to the test, metric-specific optimisation, outright fabrication — none of these are ordinary reweightings of last year's population, even when an ε amount of similar behaviour existed before. They transport mass along available action channels. Appendix B draws the reweighting-versus-transport distinction.

> **The sharp boundary.** The selection/intervention distinction is substantive exactly when agents can **move in state space** — change their (P, H) at a fixed underlying type — not merely toggle their own inclusion. If agents only choose whether to **participate** (apply, submit, enter the pool) but cannot alter their measured features, then the post-policy law is just μ restricted to the participating set and renormalised — a selection channel after all. Genuine causal Goodhart lives in features that are **cheap to change without changing the thing they were supposed to proxy**.

4.2 Response channels: the top-level object

Let $(S, \mathcal{F})$ be a measurable state space, μ_0 the baseline law, $H : S \to \mathbb{R}^d$ the hidden goal coordinates, $P : S \to \mathbb{R}$ the scalar proxy, and φ the principal's intended model $P \approx \varphi(G)$. A **response channel** is a map

$$\mathcal{R} : \Theta \to \mathcal{P}(S), \quad \theta \mapsto \mu_\theta, \quad \mu_{\theta_0} = \mu_0,$$

for some null policy θ_0. Hidden drift along the channel is $B_{H(\theta)} = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu_0}[H]$, exactly as before.

> **Definition.** Declare a type space U with baseline type law ν, a baseline kernel $K_0(ds \mid u)$, optional participation weights $W_{\theta(u)}$, and policy-indexed response kernels $K_{\theta(ds|u)}$. The induced law is
>
> $$\mu_{\theta(A)} = \left(\int W_{\theta(u)} K_{\theta(A|u)} \nu(du) \right) / \left(\int W_{\theta(u)} \nu(du) \right).$$

> The channel is **pure selection relative to $(\boldsymbol{U}, \boldsymbol{K_0})$** if $K_\theta = K_0$ for ν-almost every type and all policy dependence enters through W_θ. It is an **intervention relative to $(\boldsymbol{U}, \boldsymbol{K_0})$** if $K_\theta \neq K_0$ on a positive-mass set of types. Mutual singularity with μ_0 is decisive evidence of intervention, but not required for the causal distinction.

The entire apparatus of Chapter 2 — covariance, threshold response, weighted response — is the pure-selection case, with $L_\theta = \frac{W_\theta}{\mathbb{E}_{\mu_0}[W_\theta]}$. In Manheim and Garrabrant's taxonomy [10], **causal** Goodhart is an intervention channel in which the policy structurally breaks $P \approx \varphi(G)$, and **adversarial** Goodhart is an intervention channel in which θ is chosen worst-case for the principal. The ML instances of this regime are **strategic classification** [7] — agents manipulate features in response to a published classifier — and **performative prediction** [12] — the act of deploying a predictor changes the distribution it is predicting. The drift bound of Chapter 2 has no baseline-only analogue here: even when an intervention law is technically absolutely continuous because the baseline had ε mass on the action image, the relevant likelihood ratio reflects the induced response, not a passive selection rule. Any useful bound on intervention drift must be **imported from the agent side** — a feasibility set or cost function describing what re-arrangements of mass agents can afford. The rest of the chapter computes such a bound in toy models and reads off the controlling quantity.

4.3 A linear–Gaussian Stackelberg gaming model

A continuum of agents; an agent's type is its true quality $Q \sim \mathcal{N}\left(0, \sigma_Q^2\right)$, and the true scalar goal is $G = Q$. There is a hidden coordinate $H =$ "gaming externality": effort spent corrupting the proxy rather than improving Q — teaching to the test, metric-specific optimisation, Campbell's-law distortion [2]. The principal believes $P \approx Q$.

Baseline channel $\boldsymbol{\mu_0}$ (metric not announced, not gameable): the agent does nothing special, $P_0 = Q + \eta$ with $\eta \sim \mathcal{N}\left(0, \sigma_\eta^2\right)$ independent, and $H_0 \equiv 0$. So μ_0 is supported on the plane $\{H = 0\}$ in (Q, P, H)-space.

Intervention channel (metric announced, agents best-respond; Stackelberg, principal commits first): the principal picks a threshold policy $A_t = \{P \geq t\}$; selection is worth $V > 0$ to an agent. An agent of type Q chooses gaming effort $a \geq 0$, getting $P = Q + a + \eta$ at private cost $c(a) = a^2/(2\kappa)$ ($\kappa > 0$ measures the ease of gaming) and incurring hidden externality $H = a$. The agent maximises $V \cdot \Pr(Q + a + \eta \geq t) - a^2/(2\kappa)$.

4.3.1 The noiseless cut

Set $\sigma_\eta = 0$, so selection is $Q + a \geq t$. An agent with $Q \geq t$ sets $a = 0$. An agent with $Q < t$ either games to exactly $a = t - Q$ (cost $(t-Q)^2/(2\kappa)$, payoff V) or sets $a = 0$ (payoff 0). Gaming is worth it iff $(t-Q)^2/(2\kappa) \leq V$, i.e.

$$t - Q \leq \Delta, \qquad \Delta := \sqrt{2\kappa V}.$$

So the best response is $a^*(Q) = (t - Q) \cdot \mathbb{1}\{t - \Delta \le Q < t\}$, and the selected set is $\{Q \ge t - \Delta\}$. Compared to the **selection** counterfactual (agents cannot game; the principal thresholds $P = Q$ at the same nominal t; selected set $\{Q \ge t\}$; $H \equiv 0$):

1. **The metric lies, by up to Δ.** The principal believes selection $\Rightarrow P \ge t \Rightarrow$ (under φ) $Q \ge t$. Actually Q among the selected ranges down to $t - \Delta$. The proxy-to-goal gap $\varepsilon = P - \varphi(G) = a$ is no longer noise — it is a deliberate wedge of size up to Δ.
2. **Hidden harm appears at order Δ.** $\mathbb{E}[H \mid \text{selected}] = \mathbb{E}[(t - Q)\mathbb{1}\{t - \Delta \le Q < t\}] / \Pr(Q \ge t - \Delta) > 0$, monotone increasing in $\Delta = \sqrt{2\kappa V}$. In the selection regime H is **identically zero** in this model — consistent with Chapter 2's bound, since $s_H = 0 \Rightarrow B_H = 0$. The harm is created entirely by the intervention.
3. **The controlling quantity is not a μ_0-functional.** Δ depends on κ (technological ease of gaming) and V (the stakes), neither of which is visible in the baseline (Q, P_0, H_0) law. This is the promised contrast: intervention drift is bounded by the agents' cost–benefit ratio, not by any reweighting budget.
4. **Why this is not a selection channel.** In the clean version, μ_0 lives on $\{H = 0\}$; the post-intervention law puts positive mass on $\{H > 0\}$ (the gamers — a set of types of positive measure). The two laws are mutually singular in the (Q, H)-marginal: there is no L_θ. If the baseline instead had a small amount of pre-existing gaming variation, singularity could disappear, but the same fixed-type action response would remain. The state was **transported** along an action channel, not merely reweighted.

Proposition 2 (Stackelberg wedge). In the one-dimensional noiseless threshold model, let true quality be Q, action be $a \ge 0$, score be $Q + a$, pass condition be $Q + a \ge t$, selection value be $V > 0$, and private action cost be $a^2/(2\kappa)$ with $\kappa > 0$. Then the privately profitable gaming band has width

$$\Delta = \sqrt{2\kappa V}.$$

Agents with $Q \in [t - \Delta, t)$ choose the least passing action $a^*(Q) = t - Q$; agents below the band do not game enough to pass, and agents above threshold need no action.

Licenses: in this quadratic threshold toy, the intervention budget comes from action economics–ease of gaming and stakes–not from the baseline distribution. The metric's worst-case bias is at most Δ, and hidden action harm scales with the same band.

Does not license: a direct RLHF, finetuning, or neural-training mapping without declared analogues of a, κ, V, and the pass condition. *Toy example:* doubling the funding tied to a test score multiplies the gaming wedge by $\sqrt{2}$; halving the cost of test-prep drilling does the same.

Remark. The quadratic cost is a modelling choice. A cost with a hard cap $a \le a_{\max}$ bounds gaming at $a_{\max}$ regardless of V; a cost that is cheaper at the margin for high-Q agents skews **who** games. So $\Delta = \sqrt{2\kappa V}$ is the quadratic-cost signature, not a universal law — but the qualitative point (a positive-measure set of agents leaves the $H = 0$ locus; drift is set by agent economics, not by μ_0) is robust to the cost's shape. And if V is itself endogenous — selection is valuable only if the metric is trusted, and trust erodes as gaming is observed — there is a feedback loop not modelled here. Section 4.7 flags it.

Remark. **For ML evals.** This toy model maps cleanly to evaluator behaviour before it maps to neural training. For a benchmark designer, V is the prize attached to passing or ranking well, and κ is the ease of benchmark-specific score inflation. For a trained model, the corresponding quantity may not be a scalar cost at all: it could be an optimiser-dependent route through representation space, a memorised benchmark feature, or a reward-model vulnerability. Treating RLHF as this Stackelberg game is therefore a hypothesis to test, not a licensed conclusion of the chapter.

4.3.2 The noisy refinement (sketch)

With Gaussian proxy noise, $\Pr(\text{selected} \mid Q, a) = 1 - \Phi\big((t - Q - a)/\sigma_\eta\big)$ is smooth in a, so the best response is interior everywhere:

$$a^*(Q) = \big(\kappa V/\sigma_\eta\big) \cdot \phi\big((t - Q - a^*(Q))/\sigma_\eta\big)$$

(implicit; the right side is the marginal selection-probability gain). Every agent games a little: $a^*(Q) > 0$ for all Q, peaking near $Q \approx t - a^*$ and decaying in $|Q - (t - a^*)|$. Hidden harm $\mathbb{E}[H \mid \text{selected}] = \mathbb{E}[a^*(Q) \mid \text{selected}] > 0$ for every threshold. The hard-threshold version is the $\sigma_\eta \to 0$ limit, where gaming concentrates on the band $[t - \Delta, t)$. The lesson: noise **spreads** gaming across the whole population rather than removing it.

4.4 Multidimensional gaming: does adding a measured dimension help?

Now the dimensional thread of Chapter 1 rejoins the picture, inside the intervention regime. The motivating "deep Goodhart" move is: the principal, fighting scalar Goodhart, **adds proxy dimensions**. When agents game, does adding a measured dimension redistribute harm, conserve it, shrink it, or grow it? The answer depends critically — and instructively — on how the metric **aggregates** its dimensions, so the aggregation rule must be a visible parameter, not a hidden default. Appendices C and D give the geometric picture for the additive, conjunctive, and exchange-rate cases.

4.4.1 The exchange-rate condition

Start with the general additive case, because it is the result that survives changes of units. There are two welfare objects to keep separate. The first is fixed-deficit per-agent harm, written $H_{\text{per}(M,d)}$: conditional on an agent choosing to buy score deficit d, how much hidden harm does the cost-minimal gaming action cause? The second is population harm, written $H_{\text{pop}}\big(M, F_Q, V\big)$: after heterogeneous agents draw quality Q from F_Q and decide whether gaming is worth the prize V, how much harm is produced in the population?

Let the additive score be $\sum_{j \in M} w_j a_j$, costs $\sum_j a_j^2/(2\kappa_j)$, and hidden harm $H = \sum_j h_j a_j$. The cost-minimal allocation for a score deficit d is $a_j = d\kappa_j w_j / W_M$ with $W_M = \sum_{i \in M} \kappa_i w_i^2$, and its per-agent harm is

$$H_{\mathrm{per}(M,d)} = d \cdot \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}.$$

This is **not** invariant in M. (Two equally easy channels, $\kappa_1 = \kappa_2 = 1$, equal physical harm $h_1 = h_2 = 1$: measuring only channel 1 with $w_1 = 2$ gives $H = d/2$; measuring only channel 2 with $w_2 = 1$ gives $H = d$; measuring both gives $H = 3d/5$.) Re-routing can raise **or** lower harm, depending on the score weights. Figure 6 records the numbers.

Proposition 4 (additive exchange-rate iff). Fix a score deficit $d > 0$. For active measured channels M, suppose costs are separable quadratic, $\sum_{j \in M} a_j^2/(2\kappa_j)$ with $\kappa_j > 0$, the score is $\sum_{j \in M} w_j a_j$ with $w_j > 0$, and hidden harm is linear, $H(a) = \sum_{j \in M} h_j a_j$. The cost-minimal action satisfying the deficit is

$$a_j^* = d\kappa_j w_j / W_M, \quad W_M = \sum_{i \in M} \kappa_i w_i^2,$$

and fixed-deficit per-agent harm is

$$H_{\mathrm{per}(M,d)} = d \cdot \left(\sum_{j \in M} h_j \kappa_j w_j \right) \Big/ \left(\sum_{j \in M} \kappa_j w_j^2 \right).$$

Therefore fixed-deficit per-agent harm is conserved across active measured sets if and only if $h_j = c w_j$ on the channels being compared, in which case $H_{\mathrm{per}(M,d)} = cd$.

Licenses: under separable quadratic costs, additive score, linear harm, and a fixed score deficit, conservation of H_{per} is an exchange-rate theorem, not a dimension count slogan.

Does not license: population-level conservation, arbitrary costs, arbitrary aggregation rules, or conservation when harm-per-score ratios differ. Shared bottlenecks and correlated costs are different action geometries, not exceptions to this theorem. *Toy example:* if every point of score inflation is equally socially wasteful no matter which KPI supplies it, re-routing conserves harm; if grant-padding produces less waste per score point than citation-padding, moving weight toward grant-padding genuinely reduces harm.

Remark. This is the same structure as cost-benefit-weighted strategic classification [7], and as isoquant choice in production theory: agents pick the cheapest input bundle for a target output, while social damage is a **different** linear functional of the bundle. Conservation appears only when target output and social damage use the same exchange rates. One residual worry for later: real score weights are sometimes partly arbitrary normalisation choices, so future work must distinguish harmless unit changes from substantive incentive exchange rates.

4.4.2 Unit-weight additive model

The narrow conservation slogan is the unit-weight illustration of the exchange-rate result. Take k gaming channels and, in the base case, true quality $Q = 0$ for everyone (pure gaming). An agent allocates effort $a = (a_1, ..., a_k) \geq 0$; channel j costs $a_j^2/(2\kappa_j)$ and produces hidden harm $H_j = a_j$ (all gaming equally wasteful, $H = \sum_j a_j$). The principal **measures** a set $M \subseteq \{1, ..., k\}$ and scores additively, score $= \sum_{j \in M} a_j$ (unit weights); selection (worth V) iff score $\geq t$. An agent never games an unmeasured channel.

The agent's problem is $\min_{a_j \geq 0, j \in M} \sum_{j \in M} a_j^2/(2\kappa_j)$ subject to $\sum_{j \in M} a_j \geq t$, then game iff that minimum cost is $\leq V$. At the optimum the constraint binds and $a_j/\kappa_j = \lambda$ for all $j \in M$ (a water-filling allocation), so with $K_M := \sum_{j \in M} \kappa_j$,

$$\lambda = t/K_M, \quad a_j = t\kappa_j/K_M, \quad \text{min cost} = \lambda^2 K_M/2 = t^2/(2K_M).$$

Hence **gaming occurs iff $\boldsymbol{K_M \geq t^2/(2V)}$** (call the right side $K_{\min}$), and when it occurs, the per-agent hidden harm for this fixed pure-gaming target is $H_{\text{per}(M,t)} = \sum_{j \in M} a_j = \lambda K_M = t$ — **independent of $\boldsymbol{M}$**. Thus the earlier slogan $H = t$ is only the fixed-deficit pure-gaming object $H_{\text{per}(M,t)}$, not a population welfare claim. What M controls is (i) **whether** K_M clears $K_{\min}$, and (ii) **how** the fixed harm t is split across the measured channels — proportional to κ_j. Figure 4 shows the same water-filling geometry in effort space.

> **Conservation under re-routing (narrow form).** With a unit-weight additive metric and gaming that is equally wasteful per unit of score, the principal cannot reduce per-agent harm for a fixed pure-gaming score deficit by changing **which** channels it measures, as long as the measured set keeps enough aggregate gaming capacity ($K_M \geq K_{\min}$): $H_{\text{per}(M,t)}$ is pinned at t and merely re-routes. Closing a gamed channel just spreads t over the others. *Toy example:* a hospital ranked on readmission rates clamps down on coding of "readmission" — the gaming reappears as patient selection, discharge timing, observation-status reclassification; the per-case distortion needed to clear the cutoff is unchanged.

> **Adding a gameable measured dimension backfires at population level.** Expanding M strictly increases K_M, which **lowers** the gaming cost $t^2/(2K_M)$, which weakly **enlarges** the population that finds gaming worthwhile, even though $H_{\text{per}(M,d)} = d$ for each fixed deficit in the unit-weight equal-harm case. With quality heterogeneity $Q \sim \mathcal{N}(0, \sigma^2)$ — an agent needs score $\geq t - Q$, at cost $(t-Q)^2/(2K_M)$, and games iff $t - Q \leq \sqrt{2K_M V}$ — the gaming-eligibility cutoff $Q \geq t - \sqrt{2K_M V}$ moves **down** as K_M grows. The population object is
>
> $$H_{\text{pop}}(M, F_Q, V) = \mathbb{E}\left[H_{\text{per}(M,t-Q)} \mathbb{1}\left\{0 < t - Q \leq \sqrt{2W_M V}\right\}\right],$$

with $W_M = \sum_{j \in M} \kappa_j w_j^2$; in this unit-weight case, $W_M = K_M$ and $H_{\text{per}(M,t-Q)} = t - Q$. More agents game; the fraction of selected agents who are pure gamers rises, and H_{pop} increases whenever there is positive mass in the newly admitted gaming band. *Toy example:* a university worried that "publication count" is gamed adds "grant income" and "media mentions" to the scorecard; each is independently inflatable, so the cheapest path to any target score is now cheaper, and more faculty shift from research to portfolio-padding.

The effective levers are aggregate, not allocative. In this model the principal reduces gaming harm only by (a) shrinking aggregate gaming capacity K_M below K_{min} — narrowing the measured set to channels that are individually hard to game (small κ_j), or hardening channels; (b) raising the bar t (but this raises K_{min} too; the net effect on the $K_M \geq t^2/(2V)$ test depends on whether real signal scales with t); (c) cutting the prize V; or (d) abandoning additive aggregation. Shuffling attention between channels of equal hardness does nothing. *Toy example:* anti-cheating in exams works by making the exam itself hard to game — proctoring, item rotation — not by adding more graded components.

4.4.3 A conjunctive metric flips the sign

Suppose instead that passing requires $a_j \geq t$ for **every** $j \in M$ — the principal demands the agent clear a bar on all measured dimensions. The cost-minimal way to pass is $a_j = t$ for all $j \in M$, at cost $\sum_{j \in M} t^2/(2\kappa_j)$, and per-gamer harm $H_{\text{per}}^{\text{conj}(M,t)} = \sum_{j \in M} t = t|M|$. Now per-gamer harm grows **linearly in the number of measured dimensions**: adding a dimension unambiguously increases harm conditional on gaming — though it also raises the cost of clearing the metric, so fewer agents may do it. The sign of H_{pop} is a genuine trade-off between harm-per-gamer and gamer-count.

Gaming harm's dependence on the **number** of measured proxy dimensions is governed by the aggregation rule and by the welfare object. Unit-weight compensatory/additive metrics conserve fixed-deficit per-agent harm ($H_{\text{per}(M,t)} = t$, re-routing only) — when measured channels are equally harmful per score unit; conjunctive/min metrics **multiply** per-gamer harm ($H_{\text{per}}^{\text{conj}(M,t)} = t|M|$). Real scorecards are usually compensatory (weighted sums of KPIs), which is the regime where "just add another metric" can backfire for H_{pop} by cheapening the cheapest gaming path and expanding the gaming population. The selection-regime $\sqrt{d}$-type scaling from Chapter 2 and these intervention-regime flat/linear behaviours are **different phenomena** and should not be conflated. See Figure 5.

Remark. Scope, made explicit. The additive claims assume linear score aggregation and quadratic separable gaming costs. The unit-weight conservation illustration further assumes equal harm per score unit, a fixed pure-gaming target deficit, and the binding-constraint deterministic regime. Relax equal harm — let channel j also contribute $\gamma_j \in [0,1]$ to the true goal, so $H = \sum_j (1-\gamma_j) a_j$ — and the principal can shrink harm by steering effort onto high-γ channels. Relax fixed participation with quality heterogeneity, and aggregate population harm rises with K_M because more agents enter the gaming band. Real scorecards are often nonlinear within each KPI — saturating, threshold-clipped, rubric-scored, or capped — so the linear additive model is not innocuous. Its value is that it isolates the exchange-rate condition under which conservation is real rather than an artefact of units. The empirical discriminator is not whether a scorecard is literally quadratic; it is whether added dimensions contribute independent, substitutable gaming capacity. If the new channel is cost-correlated with existing channels, if gaming has a hard bottleneck, or if the principal dynamically reweights after seeing attacks, κ_{new} should not be counted as a full additive increment to K_M.

4.5 Evaluation-suite design implications

The selection/intervention split is directly useful for benchmarks and leaderboards. An evaluation suite should first ask what kind of pressure it creates. If it only chooses among fixed candidates, the relevant audit is a selection audit: which hidden properties covary with the reported score, how far into the tail the selection goes, and whether the weighting rule amplifies a known residual. If training or submission behaviour adapts to the eval, the audit becomes an intervention audit: what score-increasing moves are available, what they cost, what prize they buy, and which hidden harms they induce.

> **Scorecard design claim.** Adding a benchmark dimension is protective only when it either measures real capability cheaply enough to displace gaming, or raises the cost of gaming more than it raises the menu of compensatory substitutes. Under an additive leaderboard, a low-cost gameable task is an extra route to the same aggregate score. Under a conjunctive leaderboard, every task becomes a gate: substitution is blocked, but false negatives rise and any system gaming its way through must clear more bars.

This makes the usual "more metrics means less Goodhart" heuristic conditional. For an additive benchmark, the design question is whether a proposed task adds real signal or mostly adds independent κ_j; cost-correlated tasks, shared bottlenecks, hard caps, and dynamic reweighting all break the clean additive-capacity calculation. For a conjunctive benchmark, the design question is whether the extra bar screens out benchmark-specific optimisation or merely filters out genuinely capable systems with one narrow deficit.

The practical checklist is correspondingly split. For a fixed candidate pool: audit baseline tail response, hidden-score covariances, and the selection depth. For fine-tuning or repeated submissions: audit the cheapest score-increasing maneuvers, their hidden harms, the prize for passing, and whether the proposed new task is an independent route or a real bottleneck. For additive leaderboards: ask whether the new component adds signal or compensatory gaming capacity. For conjunctive gates: ask whether the added gate blocks substitution enough to justify the extra per-gamer burden and false-negative risk. Either way, eval results should be treated as experiments rather than leaderboard decorations: pre-state the comparison the score is meant to support, report uncertainty where stochasticity matters, document prompts/configs, and separate the measured score from the broader capability or alignment claim.

4.6 A convex-cost intervention bound

The selection-channel bound says that hidden drift is controlled by a chi-square/reweighting budget δ. The intervention examples say that, once agents can move in state space, the analogous budget has to come from the agents' cost geometry. In the quadratic one-dimensional model that budget is $\Delta = \sqrt{2\kappa V}$.

Proposition 3 (convex score-deficit budget). Let the fixed-type action space be finite-dimensional. Let $c : \mathbb{R}^n \to (-\infty, +\infty]$ be closed, proper, and convex, absorbing feasibility by setting $c(a) = +\infty$ outside the feasible set. Let proxy gain be linear, $p(a) = w \cdot a$. For a score deficit d, define

$$m(d) = \inf_a \{c(a) : w \cdot a \geq d\}.$$

Under standard convex-duality regularity, for example a finite-cost feasible point with $w \cdot a > d$,

$$m(d) = \sup_{\lambda \geq 0} [\lambda d - c^*(\lambda w)],$$

where $c^*(y) = \sup_a [y \cdot a - c(a)]$.

Licenses: a score-deficit/private-cost budget after the action space, proxy gain, convex cost, and regularity condition have been declared. In the single-channel quadratic case $c(a) = a^2/(2\kappa)$, this gives $m(d) = d^2/(2\kappa)$, so stakes V permit deficits $d \leq \sqrt{2\kappa V}$.

Does not license: a welfare bound without a hidden harm functional, nonconvex/fixed-charge dynamics, non-convex ML/RLHF training dynamics, or inference of the cost geometry from the baseline distribution. An ML application can import this proposition only after declaring a local action space, response model, and convex cost geometry.

The proof sketch is the standard Fenchel move. Write the Lagrangian $c(a) + \lambda(d - w \cdot a)$ with $\lambda \geq 0$. Minimizing over a gives $\lambda d - c^*(\lambda w)$, and strong duality supplies equality under the regularity condition. This is the clean convex analogue of the Stackelberg wedge: the budget is imported from declared action costs, not from μ_0.

4.7 What we have, and what is open

Chapters 2 and 3 give a clean dichotomy. **Selection channels** — the policy reweights a fixed baseline — contain all of regressional and extremal Goodhart; hidden drift is bounded, $\left\| B_{H(\theta)} \right\|_2 \leq \delta \cdot \|s\|_2$, with every term a baseline functional, and the number of dimensions enters only through $\|s\|_2$. **Intervention channels** — the policy changes fixed-type behavior through a response kernel — contain causal and adversarial Goodhart; there is no baseline-only bound, and the controlling quantity (e.g. $\Delta = \sqrt{2\kappa V}$ in quadratic Stackelberg gaming) is exogenous to μ_0, set by the responding agents' cost geometry. In the multidimensional intervention regime, "fight Goodhart by measuring more dimensions" has a precise predicted failure mode: under a compensatory rule it conserves $H_{\mathrm{per}(M,d)}$ only when channels are equally harmful per score unit, and it lowers the cheapest gaming cost, recruiting more gamers — an $H_{\mathrm{pop}}(M, F_Q, V)$ backfire; under a conjunctive rule it multiplies per-gamer harm while the population sign depends on entry. The principal's real levers are aggregate (shrink total gaming capacity, harden channels, raise the bar relative to real signal, cut the prize) or structural (pick low-harm-per-score proxies whose cheapest inflation is partly real; change weights; or go conjunctive and accept harm scaling with the number of bars).

That is the strong claim. The weaker, broader conjecture is recursive: in many real systems, once the visible failures are patched, the residual error that remains will be concentrated in dimensions that are less legible, less represented in the training or evaluation signal, or cheaper for agents to exploit. The chapters above do not prove that conjecture. They say what one would have to measure to make it precise: the baseline response curve, the coupling norm, the cost geometry, the aggregation rule, the gaming capacity, and the score-to-harm exchange rates.

Chapter 4 sharpens that last sentence. The recursive hypothesis is tempting to summarise as "residual error becomes more complex". That is not licensed. The right next object is a **response-shape prediction**: once a response channel and its constraints are specified, what shape of hidden residual should it produce?

Claim one might use	What the chapters show	Boundary
Adding additive metrics can worsen gaming.	In the deterministic quadratic water-filling model, adding an independently gameable measured channel increases K_M, lowers $t^2/(2K_M)$, and weakly expands the gaming band; this is an H_{pop} claim, not a claim that fixed-deficit H_{per} rises.	Does not cover non-substitutable, capped, correlated, or dynamically reweighted channels.
Selection and intervention are different Goodhart channels.	Pure selection reweights fixed states or types. Intervention changes the response kernel at fixed type; singularity with μ_0 is sufficient but not necessary evidence.	Absolute continuity alone does not settle causal status when the baseline already contains ε mass on the induced behaviour.
$\Delta = \sqrt{2\kappa V}$ describes neural training.	Only in the one-dimensional quadratic Stackelberg toy. It says that a future ML mapping must identify a real analogue of κ and V.	No mapping to RLHF, finetuning, or reward-model optimisation is licensed without that primitive map.
Recursive Goodhart is plausible.	The framework identifies mechanisms by which patched proxies can leave residual error in hidden dimensions.	Not a theorem; it requires empirical estimates of legibility, coupling, and gaming cost.
Goodhart drift becomes more complex.	Only after a complexity measure and response mechanism are fixed. The current results separate support, rank, description length, cost, and search accessibility.	No monotone complexity law follows from Chapters 1–4.
Minimum-complexity attractors explain recursive Goodhart.	Only conditionally: fixed-charge costs can yield sparse drift, low-rank affordances restrict drift to an image, and search priors can favour low description length.	Quadratic costs and selection channels both block the generic theorem.

Table 1: Claim audit: what these chapters license, and where the license stops.

The remaining appendices are visual. Appendices A–D illustrate the formal chapters: selection response curves, dimensional coupling, selection versus intervention, additive versus conjunctive gaming, and the exchange-rate condition for conservation. Appendix E is a speculative cartoon of the recursive-Goodhart hypothesis, explicitly not a conclusion of the formal results. Appendix F visualises the Chapter 4 response-shape repair. The research inventory — both the work-in-progress threads and the parked open questions that earlier drafts kept as separate appendices — now lives in the intro §§1.5–1.6.

5 Response shape: when hidden residuals concentrate

5.1 The repaired recursive question

The previous chapters leave a natural question. If a principal patches a visible failure, does the remaining error drift toward the "simplest" or "minimum-complexity" way to pass the proxy? This would make the recursive Goodhart picture much sharper: the system would not merely leak error into arbitrary hidden directions; it would find the easiest hidden route the principal failed to specify.

That claim is too strong as stated. "Complexity" is not one object. It can mean support size, rank, description length, entropy, KL from a reference distribution, private cost, or search accessibility. These agree in some toy cases and disagree in others. So the live question is not whether Goodhart generically selects minimum complexity. It is:

> Given a response channel, its constraints, and a pre-specified shape measure, what hidden residual shape should proxy pressure select?

This chapter gives the first answer. Selection channels follow baseline tails. Intervention channels follow cost or search geometry. A minimum-complexity attractor appears only when that geometry is aligned with the chosen complexity functional.

5.2 Why the generic attractor fails

In a selection channel, there is no optimisation over hidden residuals at all. The policy reweights a fixed baseline, and the hidden drift is still

$$B_{H(\theta)} = \mathbb{E}_{\theta[H]} - \mathbb{E}_{\mu[H]}.$$

Thus any simplicity bias in selected hidden residuals must already live in the baseline candidate distribution. A direct counterexample is diffuse: let $P = Z$ and let $H_i = Z + \xi_i$ for $i = 1, ..., d$, with independent mean-zero noise ξ_i. Thresholding on $P \geq t$ shifts every hidden coordinate by the same amount. In the substantive coordinate system $H_1, ..., H_d$, the response has full support, not minimum support.

> *Remark.* The vector $(1, ..., 1)$ may still have a short description in a symmetric representation. That is exactly the point: support-size complexity and description-length complexity are different claims.

Intervention channels give a second obstruction. Suppose an agent chooses $a \in \mathbb{R}^k$ to close a linear proxy deficit $w \cdot a \geq d$ under quadratic cost

$$c(a) = (1/2)a^T C^{-1} a,$$

with C positive definite. The KKT conditions give the unconstrained binding solution:

$$C^{-1}a = \lambda w, \quad a^* = dCw/(w^T Cw).$$

If the feasible action set also requires $a \geq 0$, this formula is an interior solution only when $Cw \geq 0$ componentwise (and the nonzero components can meet the deficit). If some component of Cw is negative, the constrained optimum is found by solving the same quadratic problem on the active feasible face, with the negative action coordinates fixed at zero or other active bounds imposed. For example, with $C = \text{diag}(1, 1)$ and $w = (1, -1)$, the unconstrained direction is proportional to $(1, -1)$, which is invalid under $a \geq 0$; the active-face solution uses $a_2 = 0$ and buys the deficit through a_1.

> **Quadratic response-shape result.** In the unconstrained quadratic model, or on a nonnegative interior face satisfying $Cw \geq 0$, proxy pressure selects the minimum-cost direction proportional to Cw. This is not a minimum-complexity direction by default. It is dense only when Cw is dense in the pre-specified action basis and no sign, cap, or other active constraints bind. *Toy example:* if all KPI-padding channels are smooth substitutes with symmetric quadratic effort costs, the cheapest way to add score spreads effort rather than using one obvious loophole.

The result is useful precisely because it blocks a rhetorical shortcut. Quadratic cost can generate diffuse drift; fixed-charge cost can generate sparse drift; search priors can generate low-description-length drift. The response process decides the shape.

5.3 A conditional response-shape taxonomy

The safe replacement for “minimum-complexity attractor” is not the empty slogan “geometry matters”. It is a table of conditional predictions:

Response geometry	Licensed prediction	Guardrail
Quadratic intervention cost	Cost-minimal drift along Cw in the unconstrained/ interior case.	For $a \geq 0$, require $Cw \geq 0$; otherwise solve on the active feasible face. Dense only when that face solution is dense.
Fixed activation or linear marginal cost	Low-support or one-channel drift in the uncapped, no-tie case.	Caps, convex post-activation costs, detection risk, and ties can force spreading.
Low-rank action map L	Hidden drift is restricted to $\mathrm{im}(L)$.	Spectral concentration requires a specified hidden representation and value basis.
Simplicity-biased search prior	Failures are biased toward low description length under that prior.	The coding language or search prior must be fixed before observing the failure.

Table 2: Response-shape predictions. The framework licenses conditional geometry-to-shape claims, not a representation-invariant law of increasing complexity.

Response-shape claim. Proxy pressure does not determine hidden drift by itself. A prediction needs a response process, a constraint set, and a pre-specified residual shape measure. Once those are fixed, the model can predict dense, sparse, low-rank, low-description-length, or goal-improving responses. Without them, "complexity increase" is only a post-hoc label.

5.4 Fixed charges, caps, and lumpy spillover

The fixed-charge row is the most concrete sparse-attractor toy. Let channels $j = 1, \ldots, k$ have action a_j, score weight $w_j > 0$, marginal cost $q_j > 0$, fixed activation cost $F_j \geq 0$, and cap $u_j \in (0, \infty]$. For a score deficit $d > 0$, the agent solves

$$\min \sum_j F_j \mathbb{1}\{a_j > 0\} + \sum_j q_j a_j$$

subject to

$$\sum_j w_j a_j \geq d, \quad 0 \leq a_j \leq u_j.$$

Write the effective marginal cost per score unit as $r_j = q_j / w_j$. The problem is infeasible iff $d > \sum_j w_j u_j$.

First, if all caps are infinite, the cost of using only channel j is $F_j + r_j d$. If the minimiser is unique, the optimum is one-channel:

$$j^*(d) \in \arg\min_j \big(F_j + r_j d\big), \quad a_{j^*} = d / w_{j^*}, \quad a_i = 0 \ \text{ for } \ i \neq j^*.$$

The active channel can change as pressure rises. Low fixed cost can win for small deficits; low marginal cost can win for large deficits.

Second, if $F_j = 0$ or activation costs are already paid, finite caps convert one-channel drift into ordered spillover. Sort channels so $r_1 < r_2 < \ldots < r_k$ and define cumulative score capacities $S_m = \sum_{j \le m} w_j u_j$. If $S_{m-1} < d \le S_m$, then channels $1, \ldots, m-1$ are full, channel m is partially used, and later channels are unused:

$$a_j = u_j \;\text{ for }\; j < m, \quad a_m = (d - S_{m-1})/w_m, \quad a_j = 0 \;\text{ for }\; j > m.$$

Third, positive fixed costs plus caps break universal sorted filling. The exact object is a finite active-set comparison: for each paid set M, pay $\sum_{j \in M} F_j$ and fill within M by increasing r_j; then choose the cheapest feasible candidate. A high-marginal, low-fixed-cost channel can win at small d, while a low-marginal, high-fixed-cost channel wins later. The optimizer may skip the small-deficit channel instead of filling it first.

> **Capped fixed-charge shape.** Uncapped linear or fixed-charge costs can produce one-channel drift. Caps convert this into lumpy spillover only in the no-activation or already-activated linear case. Positive activation costs add entry thresholds and channel switching. *Toy example:* once a visible benchmark exploit saturates or becomes detectable, additional pressure can spill into the next available exploit, but the path is a sequence of thresholded regimes rather than smooth diffusion.

5.5 What this licenses for recursive Goodhart

The recursive hypothesis remains live, but it is now better scoped. Repeated proxy repair may move residual error into dimensions that are less legible, less represented, lower-rank, cheaper to exploit, or easier to find. The framework does not say this must happen. It says what would make each version testable: pre-specified hidden axes, a response model, and a shape measure fixed before observing the failure.

> **Recursive-Goodhart license after the response-shape pass.** The chapters license mechanism-level plausibility and conditional predictions: selection follows baseline tail response; intervention follows cost/search geometry; fixed charges and caps produce lumpy support paths; low-rank affordances restrict the image of drift; search priors can bias toward low description length. They do not license a theorem that residual error generically becomes more complex over time.

6 Response modeling: what a Goodhart claim must declare

6.1 Proxy pressure is not a complete model

The preceding chapters narrow the project in a useful way. The original temptation was to look for a characteristic shape of Goodhart pressure: error spreads, concentrates, becomes more complex, or migrates into unmeasured dimensions. Each of those can happen in a model. None is forced by proxy pressure alone.

In Chapter 2, pressure only reweights a fixed baseline, so hidden drift follows the baseline response curve. In Chapter 3, pressure changes fixed-type behavior, so the relevant object is an action or cost geometry. In Chapter 4, hidden residual shape depends on the response process: quadratic costs, fixed charges, caps, low-rank affordances, and search priors make different predictions.

So the next object is methodological. A Goodhart claim should not merely say that a proxy was optimized and then an outcome changed. It should declare the response model that connects proxy pressure to target-relevant distortion.

> **Response-modeling contract.** A Goodhart claim is incomplete unless it declares the response model: what is fixed type, what can respond, what movement is feasible, what it costs, how proxy components are aggregated, what hidden target or welfare quantity is protected, and what evidence would distinguish the proposed response channel from nearby alternatives.

This is not a retreat into “everything depends on assumptions.” It is the condition under which the earlier results become usable. Once the response model is declared, the claim can point to a selection response curve, a reweighting bound, a convex intervention budget, a fixed-charge active-set comparison, an aggregation exchange-rate calculation, or a response-shape prediction. Without that declaration, “Goodhart happened” is usually too underspecified to be evidence for any particular mechanism.

This is also the transfer rule for applications. A domain such as RLHF, benchmark-driven finetuning, hospital rankings, or institutional scorecards does not inherit the algebraic bounds by analogy. It inherits the obligation to declare the type space, response channel, action/search geometry, aggregation, hidden harm, and falsifiers. If those objects match one of the toy regimes, the corresponding calculation becomes available; if they do not, the contract says which new model is missing.

6.2 The contract

The minimum contract has eight fields.

1. **Type representation.** Declare the type space U and baseline law ν. Explain why these features are treated as fixed for this comparison rather than as future responses to the policy.

2. **State and baseline behavior.** Declare the observed state space S and the baseline kernel $K_0(ds \mid u)$.
3. **Policy exposure.** State what policy, score, threshold, prize, or feedback variable θ creates pressure, and which actors observe it.
4. **Response channel.** State whether θ changes participation weights $W_{\theta(u)}$, fixed-type response kernels $K_{\theta(ds|u)}$, or both.
5. **Action geometry.** If behavior changes at fixed type, declare the action space, feasible movements, costs, caps, fixed charges, search prior, or other geometry that makes some moves available and others unavailable.
6. **Proxy and target.** State the proxy, the intended target relation $P \approx \varphi(G)$, and the hidden welfare or harm quantity whose distortion matters.
7. **Aggregation rule.** If the proxy is multidimensional, state whether components are combined additively, by weights, conjunctively, by a Pareto rule, or by some other rule.
8. **Evidence standard.** State what observation would distinguish reweighting of fixed types from fixed-type behavior change, and what observation would falsify the proposed action or cost geometry.

Each field blocks a common overclaim. Without U, selection and intervention can be redescribed after the fact. Without the action geometry, intervention has no budget. Without the aggregation rule, measured dimension count has no sign. Without a hidden harm functional, proxy movement is not yet welfare harm. Without an evidence standard, the model cannot be wrong.

6.3 Selection, intervention, and evidence

The selection/intervention split is useful precisely because it says which evidence matters. Under pure selection, policy changes only the weights on fixed type-conditional behavior. The right evidence is the baseline joint distribution, the weighting rule, the selection depth, and the hidden response curve such as $\mathbb{E}[H \mid P \geq t] - \mathbb{E}[H]$. The Chapter 2 drift bound is available because the post-policy law is a reweighting of the baseline joint law on types and states.

Under intervention, the policy changes $K_{\theta(ds|u)}$ at fixed type. Marginal distribution shifts alone usually cannot identify this. Stronger evidence comes from repeated observations of the same type before and after policy exposure, randomized or staggered exposure, action traces, exogenous variation in costs or stakes, or a structural model that makes the response geometry explicit.

The distinction is representation-relative. A type space rich enough to include the whole future response plan can make behavior change look like selection over types. A type space too coarse can make ordinary heterogeneity look like intervention. That is not a defect to hide. It is the empirical commitment the model must declare.

> **Evidence claim.** Outcome distributions can support or falsify a declared response model, but they do not by themselves identify the model. A credible Goodhart application must say what is fixed type, what is response, and what observation would separate reweighting from fixed-type behavior change.

Toy examples show the difference. If a grant agency raises a score cutoff for a fixed applicant pool and applicants do not revise their proposals, the model is pure selection. If applicants rewrite proposals after the scoring formula is announced, the same marginal score improvement may be an intervention. If a hospital ranking induces the same hospital to change coding, repeated-hospital data support a fixed-type kernel change. If only hospitals already good at coding enter the ranked population, a selection model may be adequate. A single post-policy cross-section rarely decides.

6.4 What the contract licenses

The contract turns the framework into a menu of conditional claims rather than a universal theorem.

- If the response is pure selection, use baseline response curves and reweighting bounds.
- If the response is smooth costly action, use an affordable-action set or a convex score-deficit cost.
- If the response has fixed charges or caps, expect threshold regimes, spillover, and active-set switches.
- If the proxy is multidimensional, analyze the aggregation rule and harm-per-score exchange rates.
- If the claim is about residual shape, pre-specify the response geometry and the shape measure before observing the failure.

This is why the project becomes less like a theorem saying "Goodhart has this shape" and more like a response-modeling framework. The gain is not maximal generality. The gain is that a reader can ask, for any application: what mechanism is being claimed, what calculation follows from it, and what evidence would make the claim weaker?

6.5 Application discipline

The contract is especially important for ML and institutional examples because outcome distributions invite overinterpretation. A benchmark score rise could mean selection among fixed model checkpoints, finetuning that changes behavior, contamination, prompt adaptation, tool-use strategy, or reporting changes. A school score rise could mean selecting different students, changing teaching, changing attendance, changing reporting, or real learning. These are different response models.

For an empirical or ML application, the minimum mapping should say:

- what plays the role of type;
- what plays the role of action or response;
- what cost, search, or feasibility geometry predicts the response;
- which proxy components are aggregated and how;
- which hidden outcome or welfare variable is at stake;
- what data would distinguish the proposed mechanism from selection, confounding, reporting change, or real improvement.

That mapping does not need to be perfect before the framework can be useful. It does need to be explicit enough that the claim can fail. The main error this chapter is meant to prevent is treating “the metric improved and the target did not” as enough to infer the response channel, the welfare mechanism, and the next fix.

6.6 Worked contract: MMLU benchmark pressure

MMLU is a useful worked example precisely because it is not one mechanism. Hendrycks et al. introduce it as a broad multitask accuracy benchmark across many subject areas [8]. Once the score becomes visible, the same reported improvement can come from several response channels: choosing among fixed checkpoints, repeatedly testing and adapting, finetuning on benchmark-like data, data contamination, prompt search, or optimizing a proxy reward whose improvements happen to transfer to the benchmark. Model-selection overfitting [3], adaptive holdout reuse [4], and reward-model overoptimization [5] are therefore nearby warnings, not automatic explanations.

The contract for an MMLU claim has eight fields:

- **Type.** The object is either fixed model checkpoints or model-development pipelines. A claim about selecting the best checkpoint is a selection claim. A claim about training, search, contamination, or reward optimization is a response-kernel claim.
- **Baseline behavior.** The baseline is pre-exposure answer behavior on benchmark-like questions: accuracy by subject, error type, calibration, and transfer to related but unpublished tasks.
- **Exposure.** The exposure can be the public MMLU score, reporting and leaderboard/status pressure, or an internal score used for model selection. These exposures need not induce the same response.
- **Channel.** Fixed-checkpoint selection changes weights over candidates through W_θ. Finetuning, prompt search, contamination, synthetic data filtering, and reward/proxy optimization change the behavior-generating process through K_θ.
- **Action geometry.** No action geometry is implied by the benchmark alone. It must be declared: available finetuning data, benchmark access, prompt-search budget, contamination path, reward-model loop, compute, KL penalty, or other search costs.
- **Proxy and target.** The proxy is MMLU accuracy, usually aggregated across subjects. The target might be broad multitask understanding, robustness on regenerated private tests, or a declared hidden capability vector. Without that declaration, “better MMLU” is not yet a welfare or capability claim.
- **Aggregation.** The subject/task average is a score aggregation rule. It is not automatically a welfare rule, and it does not say which subjects are substitutable for which hidden capabilities.
- **Evidence.** Relevant evidence includes same-checkpoint before/after behavior, private or regenerated tests, contamination probes, traces of prompt or training search, transfer to non-MMLU tasks, and failures that move when the benchmark format changes.

What do the propositions license here? Propositions 1 and 1′ apply only to fixed-candidate checkpoint selection with declared hidden value weights. They do not describe finetuning or contamination. Proposition 2 applies only after a one-dimensional pass threshold, stakes V, and cost parameter κ have been declared; MMLU alone supplies none of these. Proposition 3 applies only after a local convex action/search geometry has been specified. Proposition 4 applies only if benchmark components are modeled as additive channels with declared costs and hidden-harm exchange rates. Thus MMLU can instantiate the framework, but it does not by itself license Stackelberg, convex-cost, RLHF, or welfare bounds.

> **Book-level conclusion.** Goodhart behavior is not determined by proxy pressure alone. It is determined by the response channel: selection over fixed types or intervention through fixed-type action and response kernels, with shape governed by costs, caps, aggregation, affordances, search geometry, and the declared hidden welfare model.

7 Visual appendices

The appendices are visual rather than foundational. Appendices A–D illustrate claims made in the formal chapters: selection drift depends on baseline response curves, dimensional scaling requires coupling assumptions, intervention channels transport mass rather than reweight it, and adding measured dimensions changes gaming through aggregation and cost geometry. Appendix E is different: it sketches a broader recursive-Goodhart hypothesis motivated by the framework but not proved by it. The formal chapters do not show that residual error generically becomes more dimensional or more complex under repeated proxy refinement; Chapter 4 instead gives conditional response-shape predictions, and shows which quantities would have to be measured for such a claim to become precise. Appendix F illustrates those Chapter 4 predictions: quadratic cost can spread response, fixed-charge or linear cost can concentrate it, and caps plus activation costs can create lumpy spillover rather than a universal complexity increase.

8 Appendix A — Selection drift is coupling-dependent, not dimension-dependent

The selection results in Chapter 2 are deliberately conditional. A proxy threshold moves hidden coordinates through the baseline response curve $b_{H(t)} = \mathbb{E}[H \mid P \geq t] - \mathbb{E}[H]$. In the Gaussian-linear model, covariance ratios summarize this curve. Outside that model, covariance is only a local linear summary and can miss the tail response that selection actually uses.

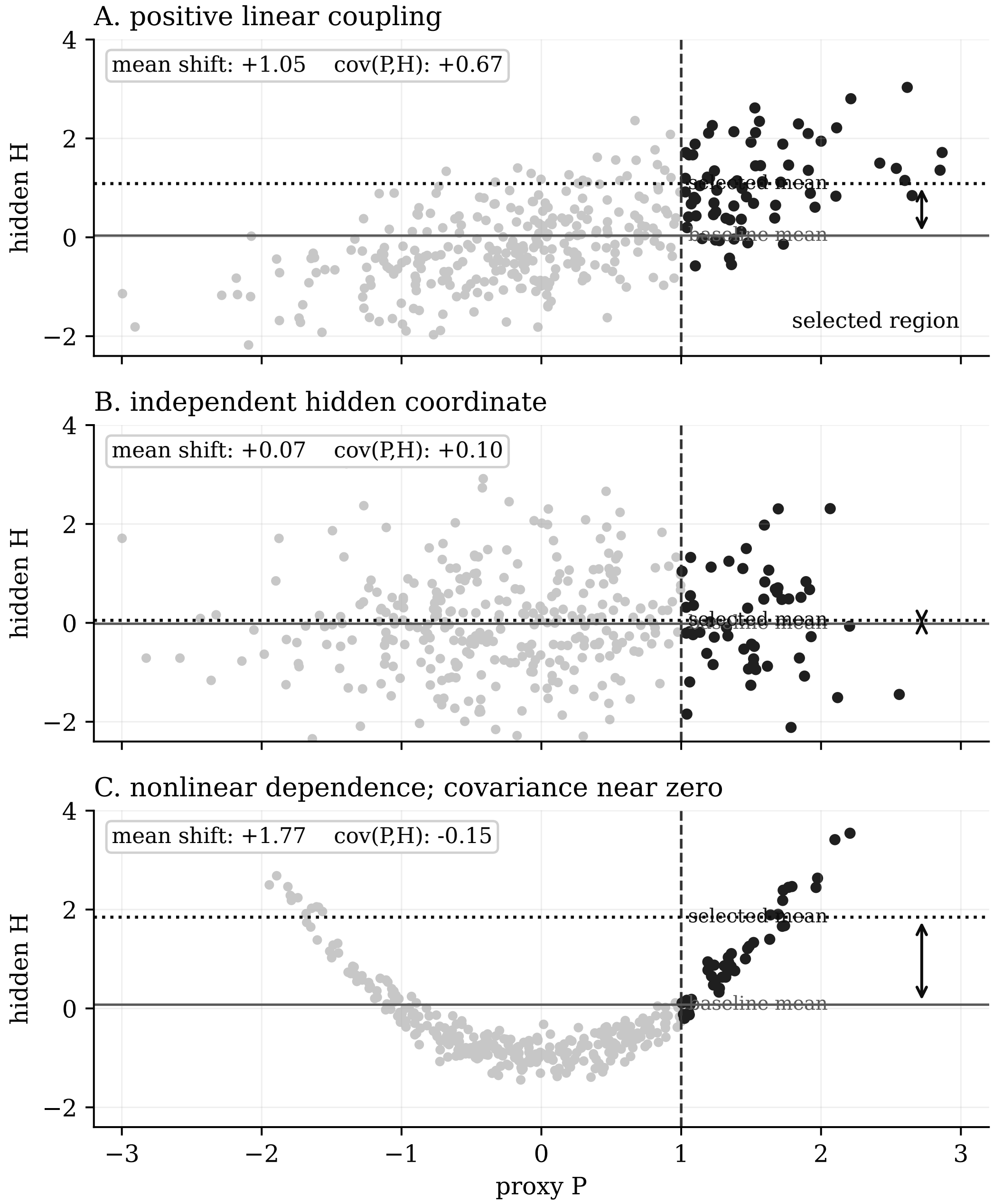

Figure 1: Selection only moves hidden dimensions through the baseline response curve. In the Gaussian-linear case, covariance summarizes this response. Outside that case, covariance can vanish while threshold response remains large. The right primitive is not baseline covariance but the selection response $\mathbb{E}[H \mid P \geq t] - \mathbb{E}[H]$.

The same point controls dimension. More hidden dimensions do not automatically create more drift. Dimensional growth appears only when adding dimensions also adds coupling to the selected proxy.

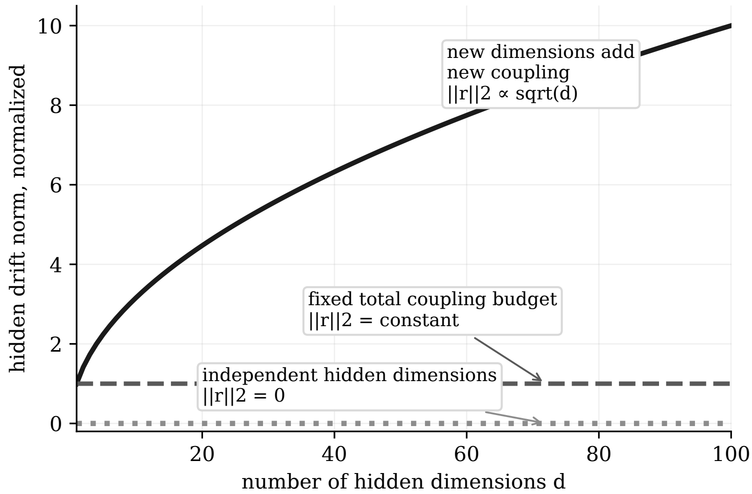

Figure 2: More hidden dimensions do not automatically imply more Goodhart drift. Dimensional scaling appears only when adding dimensions also adds coupling to the selected proxy. If total coupling is conserved, hidden drift need not grow with dimension.

9 Appendix B — Selection versus intervention as reweighting versus transport

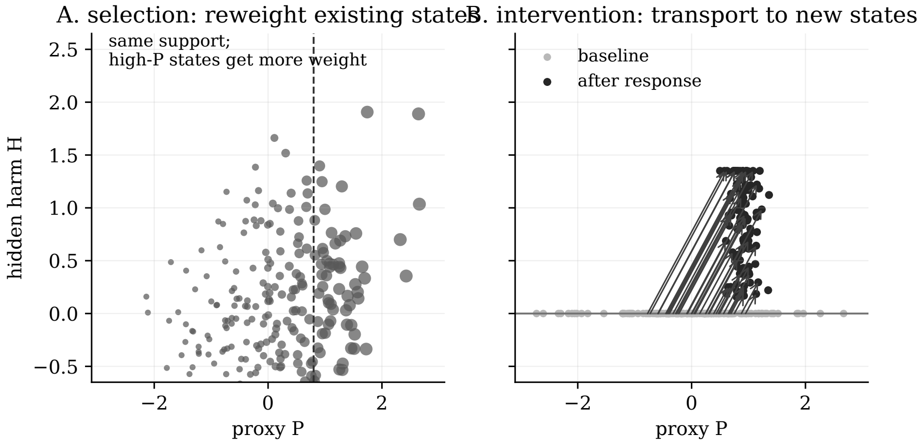

Figure 3: Selection reweights; intervention transports. A selection policy can only increase the weight of states already present in the baseline distribution. An intervention channel changes the state-generating mechanism, moving mass into regions the baseline never visited. This is why the selection-channel drift bound has no baseline-only analogue for intervention channels.

The Stackelberg toy model in Chapter 3 is the smallest algebraic version of the right panel. At baseline, $H = 0$. After the metric is announced, an agent can pay cost $a^2/(2\kappa)$ to raise the proxy by a, and selection is worth V. In the noiseless threshold case, the gaming band has width $\Delta = \sqrt{2\kappa V}$: agents with $Q \in [t - \Delta, t)$ game just enough to pass. The resulting proxy bias and hidden harm depend on κ and V, not on baseline hidden variance. Indeed, the baseline hidden variance is zero in this toy model.

10 Appendix C — Adding measured dimensions can expand the attack surface

Under additive aggregation, the agent can substitute between measured gaming channels. For unit weights and quadratic costs, the cost-minimal allocation for score threshold t is $a_j = t\kappa_j / K_M$, where $K_M = \sum_{j \in M} \kappa_j$. The minimum cost is $t^2/(2K_M)$, so gaming occurs iff $K_M \geq t^2/(2V)$. Adding a gameable measured dimension weakly increases K_M and weakly lowers the cost of reaching the same score deficit. It leaves $H_{\text{per}(M,d)} = d$ in the unit-weight equal-harm case, but can increase $H_{\text{pop}}\big(M, F_Q, V\big)$ by admitting more agents into the gaming band.

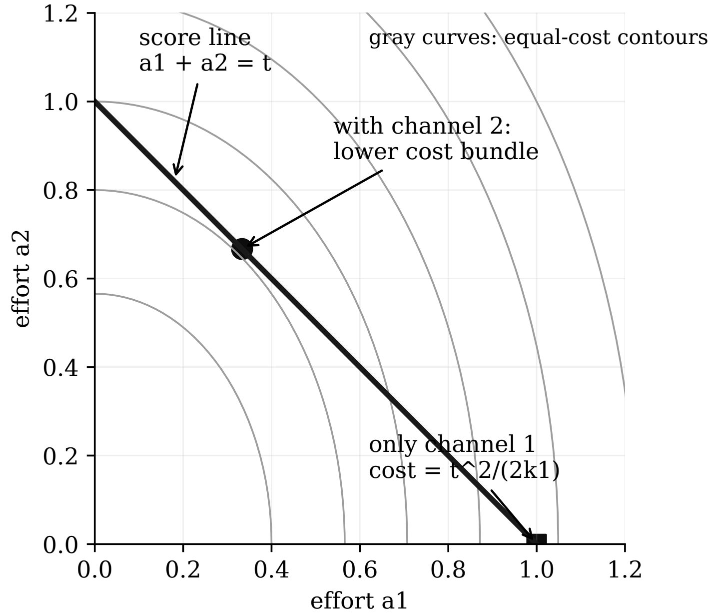

Figure 4: Under additive aggregation, adding a gameable measured dimension can lower the cost of reaching the score threshold. $H_{\text{per}(M,d)}$ may be conserved under equal harm-per-score assumptions, but the population of agents for whom gaming is worthwhile expands as aggregate gaming capacity increases.

The sign of a “more metrics” result is therefore not determined by the number of dimensions. It is determined by the aggregation rule.

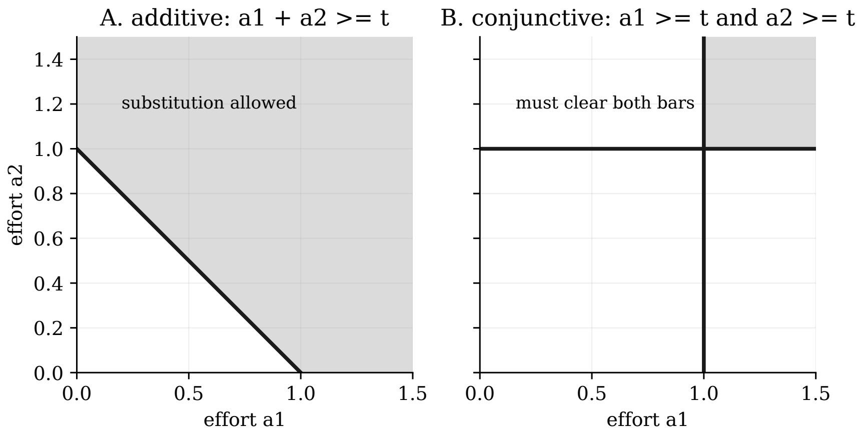

Figure 5: Aggregation rule controls dimensional effects. Additive scorecards permit substitution across dimensions; conjunctive scorecards require clearing every measured dimension. Thus adding metrics can either create cheaper routes to the same score or raise the cost of passing, depending on how the metric aggregates.

11 Appendix D — Exchange-rate condition for conservation

The narrow conservation result in the additive model assumes that a point of score inflation is equally socially harmful no matter which channel supplies it. With weighted additive score $\sum w_j a_j$, quadratic costs, and hidden harm $\sum h_j a_j$, the fixed-deficit per-agent harm is

$$H_{\text{per}(M,d)} = d \cdot \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}.$$

k1 = k2 = 1, h1 = h2 = 1, w1 = 2, w2 = 1

measured set	M	harm for deficit d
measure channel 1 only	M={1}	d/2
measure channel 2 only	M={2}	d
measure both channels	M={1,2}	3d/5

Conservation fails because score weight and social harm use different exchange rates.

Figure 6: Re-routing conserves harm only when social harm uses the same exchange rates as the score. If $h_j = cw_j$ on the active measured set, then every unit of score inflation is equally socially harmful and $H_{\text{per}(M,d)}$ is conserved. Otherwise, moving weight across channels can raise or lower fixed-deficit harm.

12 Appendix E — A speculative recursive-Goodhart cartoon

This appendix is not a theorem of Chapters 1–4. It is a cartoon of a broader empirical hypothesis suggested by the framework. The axes labelled h_1 through h_5 are deliberately not proxy dimensions. They stand for outcome-relevant properties of the policy or model that the proxy stack does not fully capture: long-horizon effects, strategic pressure, rare-context behaviour, objective stability, institutional fit, or whatever the domain-specific hidden variables turn out to be. The cartoon assigns them synthetic, mixed movements because the framework alone does not say whether their correlations with the monitored proxy dimensions are positive, negative, or close to zero.

The hypothesis would earn support if successive proxy patches predictably reduced visible residuals while increasing residuals on pre-specified hidden dimensions that were less legible, less represented in training/evaluation, or cheaper to exploit. It would lose support if hidden residuals improved along with the monitored axes, moved idiosyncratically with no relation to legibility or cost, or if the cheapest route to high score became genuinely goal-improving rather than merely less visible.

Chapter 4 supplies the guardrail for reading this cartoon. A recursive pattern should not be inferred from "complexity" after the fact. The hidden axes, the response geometry, and the relevant shape measure — support, rank, description length, cost, or search accessibility — must be specified before the patching sequence is used as evidence.

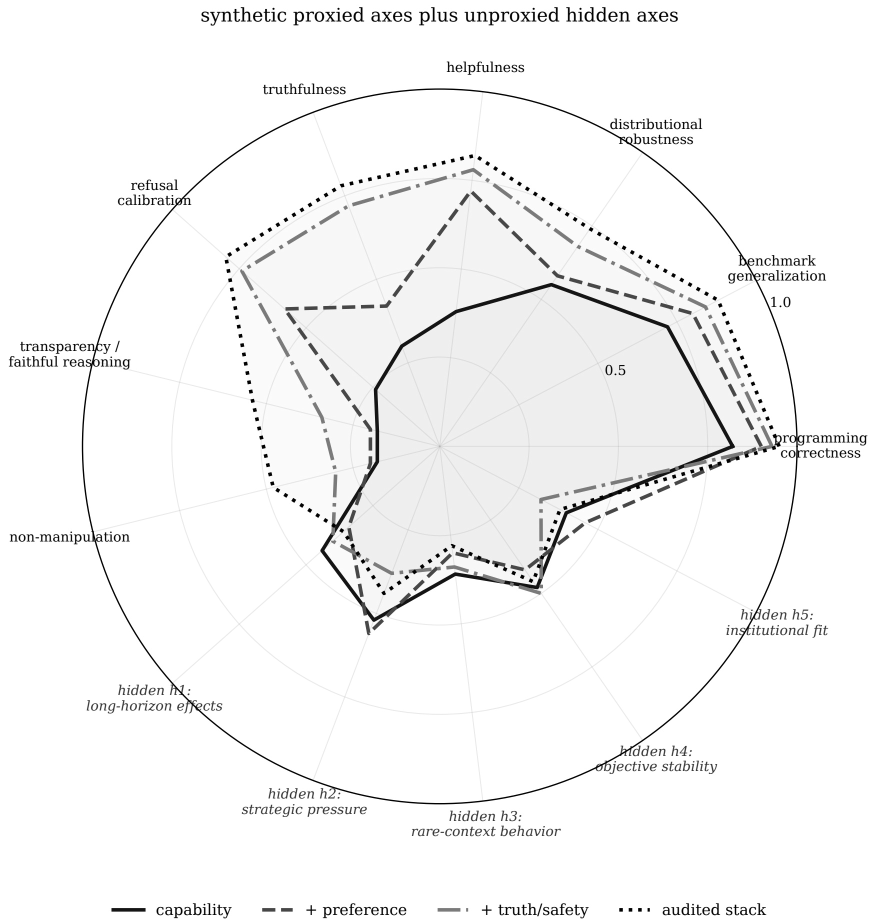

Figure 7: Speculative recursive-Goodhart cartoon. As more proxy dimensions are explicitly constrained, the cheapest route to high score changes. The monitored axes can improve while unproxied hidden dimensions $h_1, ..., h_5$ move differently, depending on their empirical correlations with the proxy stack and on the available gaming channels. In favorable cases, hidden outcome quality improves too. In unfavorable cases, residual error moves into dimensions that are less legible to the evaluator, less represented in the training signal, or cheaper for the model to exploit. The plotted values are synthetic and illustrative; they are not empirical measurements, and the hypothesis needs pre-specified hidden dimensions before it can be tested.

13 Appendix F — Response-shape predictions are conditional

Chapter 4 replaces the generic minimum-complexity attractor story with a conditional response-shape story. The relevant visual distinction is between the shape of the feasible target set and the geometry that selects one feasible response. Quadratic costs select a smooth minimum-cost direction; linear or fixed-charge costs can select one route; caps force spillover only after a route saturates.

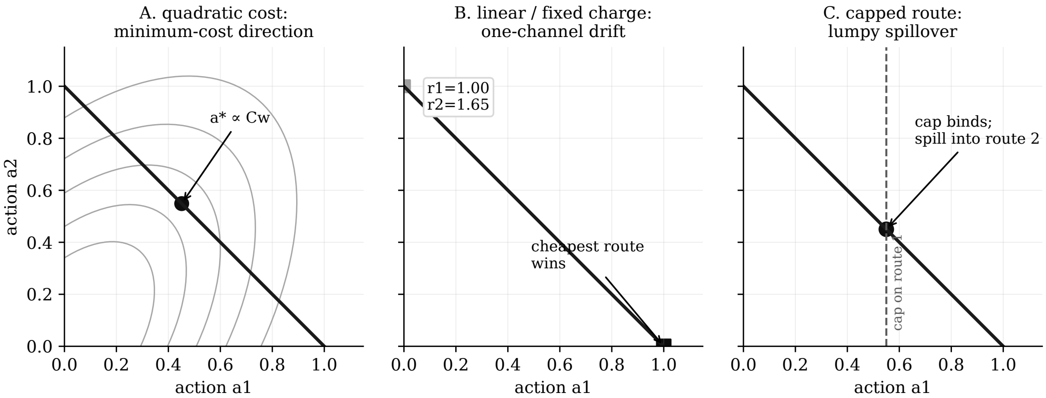

Figure 8: Response geometry selects the residual shape. A quadratic cost can spread effort along Cw; a linear or fixed-charge model can concentrate effort on the cheapest route; a cap converts that concentration into spillover. None of these is a generic law of increasing complexity.

Positive activation costs make the capped story less smooth. The optimizer may use a cheap-to-start channel for small deficits, then switch to a different channel with lower marginal cost once the deficit is large enough to justify the entry cost. Thus the robust prediction is lumpy regime change, not universal sorted filling.

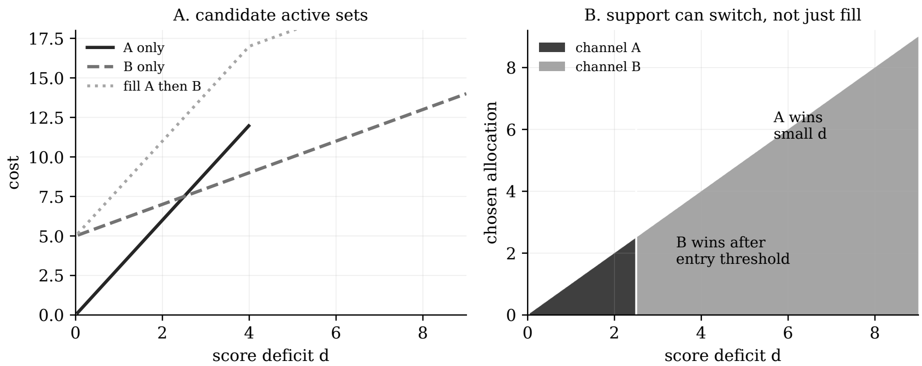

Figure 9: Fixed charges plus caps produce active-set switches. In this example, a high-marginal, zero-fixed-cost channel wins at small deficits, but a low-marginal, positive-fixed-cost channel wins later. The optimizer can skip the small-deficit route instead of filling it first.

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